

News Bias Detection During the Israel-Palestine Conflict

Shivang Kaul

DO
σ
u



Master of Science
Computer Science
School of Informatics
University of Edinburgh
2025

Abstract

The project aims to measure bias in news reporting during the Israel–Palestine conflict. A dataset of 111,678 articles from 17 English-language news sources from October 2023 to May 2025 was collected and classified for conflict relevance using keyword filtering and transformer-based models. Focusing on visual content, image-based bias detection combined computer vision and caption analysis to assess the portrayal of victims, destruction, and military actors against baselines from credible public datasets. A robustly evaluated pipeline employing NLP techniques, BLIP, CLIP, and LLMs, validated on human-coded samples, revealed significant over- and under-representation of specific actors, demonstrating a scalable approach for analysing visual bias in conflict media coverage.

Research Ethics Approval

Ethics application number: 178601

Date when approval application was submitted: 2025-04-11

This project was planned in accordance with the Informatics Research Ethics policy. It did not involve any participants; however, due to the politically sensitive nature of the topic, ethics approval was formally applied. Care has been taken to ensure that all data is handled responsibly, sources are accurately represented, and findings are presented in a neutral and objective manner.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(Shivang Kaul)

Acknowledgements

I would like to express my sincere gratitude to my supervisor, Dr. Walid Magdy, for his invaluable guidance, constructive feedback, and continuous support throughout this project. I am also grateful to Zahra Bokei, whose thoughtful comments greatly helped improve the clarity and quality of this dissertation.

I would also like to acknowledge the computing infrastructure provided by the University of Edinburgh and the University's SMASH research group, which enabled large-scale data processing and analysis for this project.

Finally, I am deeply thankful to my friends and family for their unwavering encouragement and understanding, which have been instrumental in helping me complete this work.

Table of Contents

1	Introduction	1
1.1	Research Questions and Contribution	2
1.2	Dissertation Structure	3
2	Background and Related Work	4
2.1	Brief Context on the Israel/Palestine Conflict	4
2.2	Past Studies on Analysis of Media Bias	5
2.3	NLP-Based Techniques in Bias Detection	6
2.4	Image Processing Techniques:	8
2.5	Limitations and Research Gap	9
3	Data Collection	10
3.1	Source Selection	11
3.2	Web Scraper Implementation	11
3.3	Ethical Considerations	13
3.4	Post-classification Cleaning and Enrichment	13
3.5	Storage Format and Dataset Statistics	14
4	Relevance Classification	16
4.1	Zero-shot Classification with LLMs	17
4.2	Supervised Classification	17
5	Methodology	19
5.1	Extracting Terms Specific to Israel and Palestine	20
5.1.1	Named Entity Recognition on Image Captions	20
5.2	Image Scene Detection	21
5.2.1	Caption backfilling with BLIP	22
5.2.2	CLIP based Scene Classification (Victim & Military)	22

5.2.3	Regex-based Scene Classification (Destruction)	23
5.3	Evaluation of Scene Detection	23
5.4	Attributing Images to Conflict Sides (caption-based)	24
5.4.1	Early Attempts: Rule-based and Structured NLP Approaches .	24
5.4.2	Method Used: LLM-Based Side Attribution	25
5.5	Bias Quantification	26
5.5.1	Statistical Significance Testing	27
5.6	Key Limitations	28
6	Results and Discussion	29
6.1	Military Portrayal	29
6.2	Victim Portrayal	31
6.3	Destruction Portrayal	33
6.4	Discussion	35
7	Conclusion and Future Work	39
7.1	Future Work	40
	Bibliography	41
A	First appendix	52
A.1	Code Repository	52
A.2	Data Collection JSON Format	52
A.3	Classification Prompt Template	53
A.4	NER Entity List	55
A.5	BERT and LDA Results	57
A.6	Scene Detection Classification Evaluation	58
A.7	CLIP Prompt Templates	59
A.8	Side Attribution Prompt Templates	60
A.9	Significance Tests Extended Results	62
A.10	Real-world Baselines	65
A.10.1	GED - Geo Referenced event Dataset	65
A.10.2	UN OCHA casualty dashboard	65

Chapter 1

Introduction

News bias can be defined as a portrayal of a reality that is significantly distorted, as opposed to one that differs purely due to random chance [32]. In the context of conflict reporting, this distortion can influence public opinion, shape the degree of humanitarian and political support and even contribute to the historical record of events of the said conflict[72]. The Israel-Palestine conflict is one of the most widely reported and politically polarising disputes in recent history [46]. The coverage of this conflict has often been scrutinised for not being covered equally across all outlets, owing to their political/economic affiliations and audience expectations. In reviewing how such biases are examined, much of the existing literature analyses media bias with a lens of textual analysis, analysing sentiment, topic framing, and lexical patterns [28], the role of imagery and captions has received comparatively little attention.

In the era of heightened consumption of digital short-form media that encourages frequent switching of contexts, this has been systematically analysed in studies like Chiossi's [17] and linked to reducing the ability to retain information. With this audience, visual elements in news reporting could be used to influence audience perception. Decades of psychology studies have established the "pictorial superiority effect" where images are more easily remembered as compared to their similar verbal descriptions [48]. These findings sum up the importance of images as tools to capture and influence audience perception rather than just being a supplementary medium for news distribution. This study operationalises bias as outlet-level deviation of visual portrayals from independent baselines portraying the ground truth and cross-outlet average to compare the visual portrayal of Israelis or Palestinians as aggressors or victims.

Despite the demonstrated significance of the visual medium in influencing audience perception, most systematic research at scale has been focused on the textual medium

thus, visual bias remains under-studied at this scale. Earlier studies have analysed sentiment analysis, topic framing and lexical choices [8, 32] however, large-scale computational analysis of visual media specific to this conflict remains limited. This **study fills the gap** by delivering a scalable pipeline for image and caption analysis, combining image analysis with comprehensive event baselines to quantify and observe the deviation in visual portrayal across different news sources. The study focused on covering the conflict from the start of active conflict, i.e October 2023, to May 2025 across the world's leading English language news outlets by audience count.

Analysing this dataset through a combination of image and textual lenses offers a comprehensive view, laying the foundation for addressing the research questions outlined below. This dissertation builds upon an earlier dataset collected by undergraduate researchers, extending its scope with additional scraping, cleaning, and image caption pairing. It focuses on the visual and image captions dimensions of bias, while a research colleague conducted a parallel analysis focusing more deeply on textual bias, providing a complementary perspective on the same dataset.

1.1 Research Questions and Contribution

This study examines the published news article corpus by creating a targeted corpus of multi-modal articles and then formulating a framework for analysing the bias in the coverage of the conflict across the sources. It addresses the following research questions:

- **RQ1:** How can the portrayals of Israeli and Palestinian actors be quantified to measure bias across news outlets?
- **RQ2:** How is bias manifested in visual content, particularly in victim portrayal, destruction depiction, and military representation, and how do these patterns vary across sources?
- **RQ3:** How do these patterns compare to independent conflict event baselines, and are the differences statistically significant?

The *novelty* of this study lies in both its scale and systematic analysis techniques. It collects a corpus of over 111,678 keyword-filtered articles spanning over 17 news outlets and 2.5 years, and further refines it to around 42,000 relevant articles pertaining specifically to this conflict. Each article is paired with its associated images and

captions. The analysis employs an integrated classification, natural language processing and computer vision pipeline that utilises LLMs, object detection and statistical analysis to analyse bias.

Building on these, the outcomes of the study include a comparative quantification of bias metrics across the chosen digital news outlets, identifying disparities in the portrayal of Israeli or Palestinian actors as victims or aggressors in conflict reporting. These disparities are systematically measured against actual event data using independent conflict event baselines to assess deviations from real-world event proportions. This produces a corpus of conflict-related articles, aggregate statistics and outlet-specific analysis showing how bias may manifest in the *visual* coverage of the conflict.

1.2 Dissertation Structure

The remainder of the dissertation is organised as follows. Chapter 2 gives a brief background of the conflict and reviews the current literature on media bias, media bias on this conflict and multi-modal bias analysis techniques. Chapter 3 discusses the data collection effort, along with Chapter 4 outlining the classification of relevant articles, data pre-processing and the contribution of other researchers in building the complete corpus. Chapter 5 dives deeper into the methodology for creating the pipeline for image analysis and side attribution. Chapter 6 presents the results, focusing on the portrayal of victims, destruction, and military actors, alongside statistical comparisons with independent baselines and further discusses the implications and interpretation of the findings, along with stating the study's limitations. Finally, Chapter 7 concludes with a summary of findings and directions for future research.

Chapter 2

Background and Related Work

2.1 Brief Context on the Israel/Palestine Conflict

The Israel-Palestine conflict has its roots in the Jewish migration into the Ottoman-controlled Palestine region in the late 19th century [73, 47]. In 1917, Britain assumed control of the territory post World War 1 and established a Jewish homeland in the region through the 1917 Balfour Declaration [52], which excluded the native Palestinian Arabs' political rights. This shift altered the demographic and political balance in the region, intensifying tensions between the two communities [67]. Since then, the dispute between the Jewish and Palestinian Arab national identities has caused numerous conflicts, including the 1948 Arab-Israeli War [31], the 1967 Six-Day War, the Oslo Accords, and the First & Second Intifada, which worsened the territorial disputes and mutual distrust [25]. More recently, on October 7 2023, the Hamas-led attacks on Southern Israel resulted in large-scale casualties, prompting large-scale military operations by the Israeli military in Gaza [1], sparking the extended conflict period that is central to this study.

The conflict's duration, geo-political stakes and humanitarian consequences have made it a prime focus on media research with lexical framing research such as Foster's [68] study on using terms like "killed" vs "died" or Lara Gibson's linguistic analysis [2] on use of Active vs Passive voice for Israeli or Palestinian actors/victims selectively. Mainstream Western media outlets have often been criticised for coverage bias by several studies. Differences in editorial policy cause differing portrayals or coverages of these entities that can be examined to comment on media bias [37, 38]. In recent times, the reporting media have shifted to include more images to amplify the emotional impact of the stories and thus, created new avenues for media bias research on the basis

of selective imagery or visual framing[55]. The period since October 2023 is marked by active conflict, heightened military activity and high civilian casualties, providing a rich context for examining media bias trends evolution in times of conflict.

2.2 Past Studies on Analysis of Media Bias

Foundational Studies: Media bias as a whole has been examined for decades. Studies like Groeling’s [32] have paved the way by identifying core challenges in bias quantification, like choosing baselines for comparison across outlets, and Entman[22] showed how framing disparity can influence public interpretation and shape political decisions. Building on such frameworks, Spinde et al. [70] proposed a media bias taxonomy reviewing 3000+ papers on automated bias detection showing the promise with NLP and transformer-based methods; however, they proposed that the current literature is **limited** since it lacks a multi-modal study of such trends over a longer period of time, which forms a grounding for the present study. Moreover the studies mainly either deal with media bias conceptually or the analysis is manual and thus not done at scale.

Studies on the Israel-Palestine Conflict: Foundational works like Deprez & Raeymaeckers’ [20] employed manual methods to systematically show bias towards Israeli perspectives in Flemish papers during the First and Second Intifadas. This helped establish the existence of biased reporting; however was **limited** since manual analysis of rapidly changing events posed the challenge of scalability. One of the first attempts to computationally analyse bias in conflicts was taken by Al-Sarraj & Lubbad [6] using automated SVM analysis of lexicons to establish pro-Israeli bias in Western media. However, **key limitations** were that fixed lexicons miss contextual nuance like sarcasm, etc, which may be better captured by deep learning methods. Further studies by Darwish and Magdy et al. extended this analysis to social media content. Darwish[18] showed how Twitter (now X) users, during the Gaza 2021 conflict, tend to interact with content, bolstering their beliefs, thus creating echo chambers. Magdy et al. [45] showed how platforms like Facebook may moderate content disproportionately, which may be interpreted as platform-level bias. Both these studies show us how algorithms, policies and user behaviour can influence the visibility of content in online media but could be extended to analyse visual cues as well.

Studies on Other Conflicts: Comparing studies done on parallel conflicts offers further insights. Pfau et al.[53] documented pro-military framing by reporters embedded with the military since they depended on them for support & security. Moreover, studies

like Pantti’s [50] and Ojala et al.’s [49] reiterate the importance of visual image bias in conflict reporting during the Syrian and Ukrainian conflicts, respectively. A key **limitation** to note in this group would be that both these studies on visual analysis analyse the visual frames qualitatively but don’t anchor their findings against an event-based or objective baseline.

Collectively, this literature shows that bias in conflict reporting manifests across textual and visual modalities, shaped by editorial choices, access constraints, and platform dynamics. It also underscores the need for scalable, multimodal, and temporally grounded approaches, principles that underpin the methodology of the present study. To place this study within existing computational approaches for visual analysis, this section briefly reviews various NLP and image processing techniques that could be adapted for bias analysis.

2.3 NLP-Based Techniques in Bias Detection

Early computational studies, such as the one conducted by Recasens et al.[60], focused on lexical analysis or heuristic rules to identify specific terms or phrases to identify bias. These methods were useful to identify bias but were **limited** in their ability to catch nuanced bias. Studies like Baly et al’s[9] advanced this field by incorporating transformer-based methods for identifying bias trends. Hamborg et al.[33] surveyed various techniques, showing that automated methods like word embeddings can, in fact, capture nuanced linguistic bias. Baly et al’s study was mainly trained on Western data, raising questions of cross-domain reproducibility; however, these studies show promise that computational bias detection at scale can be addressed by such approaches.

Classification Methods for Bias Relevance have evolved from linear models to transformer-based models as compared in Baly et al’s [9] study. Their multi-task setup achieved a 7–9 point F1 score improvement over n-gram and lexicon-based baselines, demonstrating transformers’ capacity for nuanced bias detection over linear models. RoBERTa[43] refines BERT’s [21] architecture, thus resulting in higher accuracy, but that comes at a cost of more extensive training and higher compute costs for inference. More recent work has explored the performance of Large Language Models like Qwen-235B [84] for zero-shot classification of data; however, the modest gains in accuracy observed in some settings may not offset the higher compute requirements and latency at inference time, especially for large-scale deployments. A practical compromise adopted in some studies is to use such models for high-quality annotation of smaller

datasets, which can then be used to fine-tune lighter transformer models like BERT for efficient large-scale inference[7].

Traditional Bias Computation Methods focus on framing and sentiment analysis, like Gilbert and Hutto’s [36] study using VADER. VADER is fast and interpretable but depends on predefined heuristics, making it brittle in the face of linguistic devices such as irony, euphemism or negation. Moreover, sentiment analysis may not be appropriate for conflict reporting since most of the articles have a negative sentiment regardless of the stance. Topic modelling approaches like Latent Dirichlet Allocation[13] have shown promise to extract thematic frames like in Verbytska’s [81] study of Russia-Ukraine conflict coverage, but have shown to overlook contextual cues since they treat documents as unordered term distributions. These features may be captured better using word embeddings as demonstrated in Spinde’s study on identifying bias terms from outlet-trained word embedding models[71]. However, any of these features need to be tagged to specific sides in the conflict to make the results interpretable, motivating the use of entity recognition and role labelling techniques discussed next.

Entity and Role Extraction: Bias in conflict hinges on extracting entities to answer questions like ”Who did what?” Named Entity Recognition (NER)[83] serves as the first step, identifying key actors such as “Israel” and “Palestine” in text, but alone does not capture their narrative roles. For e.g., linking IDF to Israel or Hamas to Palestine. Semantic Role Labelling (SRL) with BERT models and AllenNLP’s [27] implementation has been shown to accurately extract subject, predicate and verbs from long form sentences, but degrade sharply on short form texts as is established in Gardner’s [27] paper, since they are trained on well-formed, complete sentences. A parallel approach is using Transformer Q&A models such as DistilBERT[62] fine-tuned on SQuAD[58] along with NER. This approach can effectively answer well-formed questions, but thorough evaluation and fine-tuning may be required to tackle implicit roles or those requiring cross-sentence inference. Consequently, Large Language Models like GPT 3.5[16], Claude Opus have been shown to identify implicit roles accurately and are accurate for zero-shot actor tagging, but incur the highest inference cost amongst all the approaches.

Overall, entity and role extraction provides the necessary structural foundation for interpreting bias indicators in context, and when combined with visual analysis in the following section, enables a richer, multimodal understanding of how conflict narratives are framed.

2.4 Image Processing Techniques:

As seen in the literature till now, textual bias has been reviewed and analysed at length, but visual bias framing and analyses have been limited in comparison. However, the studies mentioned in section 2.1 show how it is gaining traction. Especially in conflict reporting, images can shape opinion by selective portrayal of actors or direction of violence. This often reinforces the narrative of the article. Early research on this topic, like Deprez and Raeymaeckers' [20] study, relied on manual coding and, as mentioned earlier, is limited in scalability. However, the study was instrumental at the time to demonstrate the presence of bias in the visual channel.

Automated Object Detection has been one of the techniques that have been used to identify visual indicators of bias and convert them to a textual format for easier analysis. YOLO[61] and fast R-CNN[30] remain leaders in this field as established in their seminal studies. YOLO is generally well-suited for large-scale object detection of civilians or weapons in images. R-CNN is often more accurate but being a deep learning model, it needs intensive computing to be fine-tuned able to process large amounts of data. However, these models just detect objects and need the context [66] often supported by NLP methods to be able to attribute these objects to favour or oppose a specific side in the conflict.

Again, circling back to the **transformer-based models**, BLIP and CLIP models used for different functions can be used to fill this gap. CLIP is a joint vision model trained on 400 million image-text pairs [57]. This enables it to be very effective for fast zero-shot queries like "Does the image show a destroyed building?". CLIP has been shown to provide string accuracy metrics against widely regarded image benchmarks like Flickr30k[54] and MS-COCO [42], achieving 76.2% top-1 accuracy on ImageNet zero-shot classification without fine-tuning. For the conflict reporting domain, Vongala et al. [82] established CLIP's accuracy to be close to 86% for augmented simple image and text matching, like identifying military or victims in images against industry image benchmarks making it suitable for understanding image nature that can help detect bias in framing victims versus military actors. However, performance drops are seen in its use in more complex tasks that require fine-tuning. Similarly, the BLIP model [40] is used for generating natural language captions from images. It has been established to be quite accurate, achieving CIDEr [80] scores of 127.6 on MS-COCO, competitive with state-of-the-art captioning models. Using the BLIP model, in the seminal study, Li et al. [40] have shown that the model's caption enrichment can increase the retrieval recall by

10-15%. Since both of these models need little to no training, they have increasingly been used to analyse large-scale image datasets at scale.

Other computer vision techniques, like CNN-based classifiers, have shown promise in complicated tasks like detecting emotions like "grief" or "aggression" from images, similar to the emotion tagging study by Agung et al. [4]. However, these models require large domain-specific annotated datasets for training and significant computing resources for inference. In addition, as shown by studies like Calderon et al. [15], these fine-tuned models generally do not reproduce the same accuracy metrics across domains without re-tuning. Thus, such models need to be fine-tuned for each task with balanced annotated datasets.

2.5 Limitations and Research Gap

The literature reviewed thus far demonstrates various studies on bias identification and NLP and image processing techniques that have been used to derive interpretations of bias across multiple studies. However, studies applying this multi-modal analysis technique across large-scale data on the Israel-Palestine conflict are still an area that has scope for further research. Text focus worked like Baly [9] and Hamborg [33] have analysed textual bias separately. Studies on visual bias show promise using manual techniques, but automated methods show promise for large-scale applications. In general, multi-modal frameworks like Kim et al.'s [39] GENIUS framework have shown promise on retrieval tasks, but their application in conflict-related media bias is yet to be explored.

Further, with the advancements in AI, the adoption of zero-shot methods like BLIP, CLIP, and LLMs can be utilised in journalism-based bias studies to analyse bias trends and scalable annotation for further model development. This study aims to address these gaps by systematically building a corpus of conflict-related articles across outlets focused on the Israel-Palestine conflict. It then utilised these zero-shot methods to critically analyse images, their associated captions, and full article text to quantify patterns of visual framing, actor portrayal, and lexical drift over time, enabling a multimodal understanding of how bias manifests and evolves during periods of sustained conflict.

Chapter 3

Data Collection

This chapter provides a brief outline of the data collection strategy. A multi-outlet (17 outlets), multi-modal corpus was created by collecting news articles regarding the Israel-Palestine conflict. The chosen study window is from October 2023 to May 2025, this was chosen from the date of start of active conflict post the 7th October attacks till May 2025 was chosen to maximise article coverage since the study was conducted in June 2025. This multi-modal data corpus spanning 111,678 articles accompanied by images and caption metadata is one of the **key contributions** from this study making it, to our knowledge (verified Aug 2025), **among the largest openly documented, purpose-built datasets focused on the 2023–25 Israel–Gaza conflict**; other similar datasets reviewed were limited in scale such as the Washington Institute’s Gaza war media database covering 436 stories from seven U.S. outlets [75], or are general-purpose multimodal corpora not conflict-scoped, such as GoodNews [12] and NYTimes800k [76]. Bias-detection shared tasks like FIGNEWS [19] also exist, but these focus on short posts rather than large-scale article+image coverage. Along with the corpus, the maintainable and reusable crawlers enable other researchers to re-use this corpus or re-use the crawlers to scrape these sources for new keywords.

The **work was split** between me and my colleague to scrape these outlets. The crawlers used were built upon existing work of undergraduate researchers who had used the crawlers to crawl these outlets for textual data for the period of Oct 2023 - Oct 2024. I updated the crawlers and configurations to re-run the crawlers for 9 outlets adding functionalities to scrape images and captions as well and extended the study period to Oct 2023 to May 2025. My colleague did the same for the remaining 9 sources to collectively create a corpus of 18 outlets (including ”The Sun” which was dropped from the image analysis due to low image coverage).

3.1 Source Selection

For the purpose of the study only English language sources were selected based on global audience consumption as reported by the Press Gazette (2025) [56] based on web traffic. Moreover, only the news sources which allow ethical scraping based on their robots.txt and are free to read were considered. The following 18 news sources were included: *ABC News*, *Indian Express*, *The Guardian*, *The Independent*, *News 18*, *Express (UK)*, *CNN*, *AP News*, *India(.com)*, *BBC*, *USA Today*, *The Sun*, *Newsweek*, *Hindustan Times*, *Daily Mail*, *New York Post*, *Fox News*, and *CNBC*. However, *The Sun* had no usable images in Oct 2023–Dec 2024 and was excluded, leaving 17 analysed outlets.

The following sources were excluded from the study, along with reasons specified: **Yahoo Finance**, **People**, **India Times** Domain-specific finance, entertainment news websites generally do not cover conflict stories. **Google News**, **MSN**, **Yahoo News** These websites are news aggregators, publishing news that's not their own.

Al-Jazeera Robots.txt discourages scraping for all reasons.

Forbes, **New York Times**, **Washington Post** Paid websites, news is not available to scrape or read for free.

Pipeline staging To control volume and cost, relevance classification was run on the scraped data post minimal pre-processing (Chapter 4), and then cleaning and enrichment (date normalisation and image downloads) was performed only on the subset of articles relevant to this study. All raw HTML and minimal metadata were retained for the full crawl to preserve reproducibility.

3.2 Web Scraper Implementation

As mentioned above, the earlier versions of spider crawler created by the undergraduate researchers were updated. These spiders were built using the python Scrapy library [65]. To get the URLs to be scraped for each source, sitemaps were used. Each outlet generally stores date partitioned list of sitemaps having a URL list of articles published on that day or month. For most of the outlets, sitemap links were used to drive the spiders. In the case of AP News, some articles were loaded dynamically using JavaScript so in this case the complete XML sitemaps were downloaded from their website. In both cases, the date filter was used to only include sitemap pages within the study's date range (Oct 2023 to May 2025). To control the volume of articles collected, a keyword

filter was applied on the scanned article's headline and article text. Only the articles mentioning the keywords were scraped: Palestine, Palestinian(s), Israel, Israeli(s), Gaza, Hamas, IDF / Israeli Defence Force(s) / Israeli Defense Force(s), October 7 (including variant spellings and date formats).

For extracting the images and captions from within the articles, a deterministic approach was used. Each website's sample pages were inspected manually for the article image HTML templates and how it differed from the images that were outside the article, for e.g. images of the suggested articles or advertisements. Then this sample configuration was run on a subset of 100 URLs to see if the extracted images and captions are relevant and then the spider was run for the complete sitemap range. Finally the extracted articles were stored in a standard JSON format (see appendix A.2) and each spider run was logged for maintainability.

Listing 3.1: Sample extraction config for BBC

```
SITE_CONFIG = {
    # article fields
    'title_path': '//h1[contains(@id,"main-heading")]_or_
        contains(@class,"Heading")]_or_contains(@class,"bWszMR")
        ]//text()',
    'text_path': '//div[contains(@class,"
        RichTextComponentWrapper")]_or_contains(@data-component
        ,"text-block")]//p//text()',
    'date_path': '//time/@datetime_|_//time[not(@datetime)]_
        and_contains(@class,"fkLXLN")]//text()',
    'author_path': '//div[contains(@class,"bpnWmT")]_|_//div[
        contains(@class,"TextContributorName")]//text()',
    # images & captions
    'image_path': '//div[@data-component="image-block"]//img/
        @src_|_//figure//img/@src',
    'caption_path': '//figcaption//p//text()_|_//figure//
        figcaption//text()',
    # (optional) sitemap patterns, e.g. monthly partitions
    'sitemap_index': 'https://example.com/sitemap_index.xml',
    'sitemap_pattern': r'https://example.com/sitemaps/news-(\d
        {4})-(\d{2}).xml'
}
```

3.3 Ethical Considerations

Before finalising the sources for data collection, each outlet's robots.txt was reviewed to ensure that such scraping is allowed and only then the source was selected for data collection. Some outlets mentioned list of URLs to avoid for scrapers for e.g. user keyword search pages, we respected these considerations by only limiting our scrapers to access the URLs using sitemaps. . We did not circumvent paywalls, authentication, or technical measures, and we limited ourselves to publicly accessible article pages. To ensure the servers are not overloaded during scraping, the scraping batches used minimum workers (less than 5). Conservative throttling was applied to add inter-request delays and periodic waits after every 100 requests. As a result, some site crawls like BBC ran over multiple days, this was intentional to prioritise ethical collection over speed.

3.4 Post-classification Cleaning and Enrichment

Following relevance classification (Chapter 4), basic cleaning steps were implemented for the relevant articles.

Text. Irrelevant blocks like (subscription prompts, related-story widgets, “read more” blocks) were removed using regular expressions. White space, extra quotes and newline characters were also removed.

Dates. Timestamps were parsed to UTC ISO 8601 into `date_published`. In some cases, the spiders could not pick the dates, in these cases the dates were backfilled using the article URLs which contained the published dates in most cases. All the dates were standardised to a set date format without time-zones.

Deduplication. Since our work involved scraping free text articles, the article URL was treated as the identifier field. Using the article URL, duplicates were removed while comparing the article content and images and the article with most data was persisted.

Image Downloads. An automated script was written to pick the image URLs scraped for each relevant article and download the images to our local server. Also, a new field for `local_img_path` was added to each image entry so keep track of the downloaded images and link them to corresponding articles. This was used to enable quicker image analysis by downloading a corpus of more than 100,000 images across 42,000 relevant articles. In some cases, articles contained repetitive carousels of images, these were extracted and placed in a separate JSON file so that the image analysis is only run on

the article images.

Limitation (staging bias). Since the image download, cleaning and date enrichment was done only on the relevant articles. These pre/post classification counts are reported in the Chapter 4.

3.5 Storage Format and Dataset Statistics

Normalised JSON records were converted to Apache **Parquet** for storage. Parquet’s columnar layout and compression helped store the data utilising lesser space. Also, parquet is easier to access and reduces the chances of accidental data corruption.

Final schema. { title, description, text, url, source_domain, date_published, authors, matched_keywords, keywords, images: [{src, caption, credit}], id, predicted, predicted_label, captions, lang, label_index, label_probs }.

Figure 3.1 visualises weekly article counts. Pre-/post-classification comparisons are provided in Chapter 4.

Outlet	Articles
abcnews.go.com	1705
apnews.com	10416
bbc.com*	4848
cnbc.com	1958
cnn.com*	4190
dailymail.co.uk	10746
express.co.uk	5805
foxnews.com	11969
hindustantimes.com*	6619
india.com*	1326
indianexpress.com*	6291
independent.co.uk	14137
news18.com*	4098
newsweek.com*	7778
nypost.com*	11106
theguardian.com	5028
thesun.co.uk	498
usatoday.com*	3160
Total	111,678

Table 3.1: Per-source article counts (Oct 2023–May 2025) – Pre-classification. Sources marked with * in green were scraped by me.

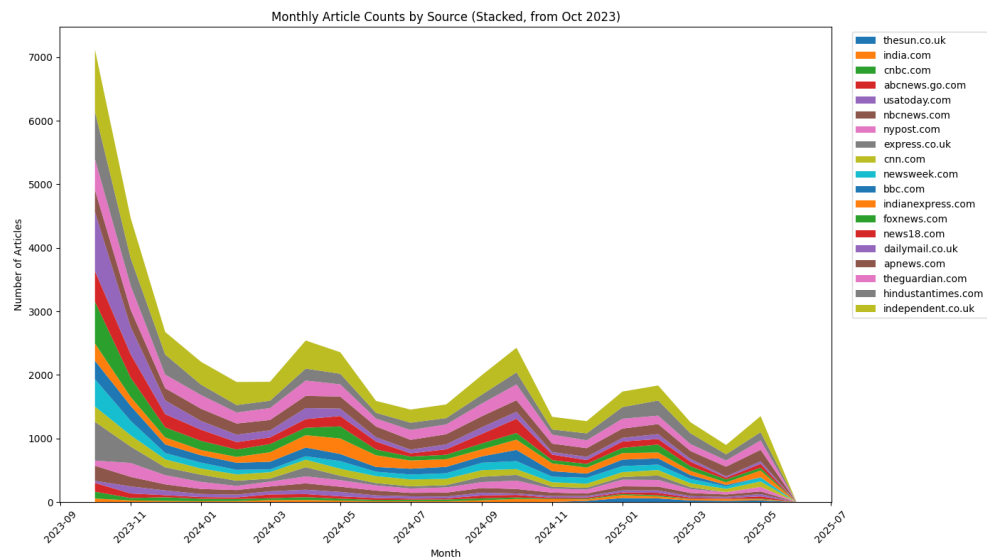


Figure 3.1: Monthly article counts across all outlets (Oct 2023–May 2025; post-classification subset).

Publish Count Analysis Observations It is observed that some outlets, like The Independent and The Guardian, publish more articles as compared to others. In the Figure 3.1, it can be seen that the conflict reporting cycles overlap and almost all spikes occur in parallel for all sources. These periods of increased reporting are as follows:

October 2023 Largest spike corresponding to the initial escalation following the 7th October attacks.

April/May 2024 Coverage rose as Israel intensified strikes in Nuseirat and eastern Rafah, attracting international diplomacy talks amid the backdrop of the Eid festival. [10]

October 2024 With the active conflict reaching its one-year mark, a lot of reporting involved a timeline and analysis of the conflict and how it grew over the year. [3, 5]

February 2025 Increases reporting on the ceasefire agreement introduced on Jan 17, 2025 [79]

Late April 2025 Smaller spike possibly due to increasing calls for a stable ceasefire via international channels and its related developments. [74]

Reproducibility Crawler code and site scraper spiders were versioned and the configurations used for scraping were saved and logged. The raw HTML parsed as JSON was saved to enable reproducibility and guide future researchers to utilise the same for similar data collection tasks.

Chapter 4

Relevance Classification

To study framing and bias indicators, a domain-specific dataset is essential. For our study we needed to filter out the relevant articles reporting only on the Israel-Palestine conflict to weed out noise due to other non-focussed articles. The initial data collection was done based on keywords however, articles that talk about Israel's trade or finances also show up which may skew our bias results. Thus, this chapter explains how relevance classification was done on the corpus to create a domain-specific dataset for this study.

Contributions: This chapter summarises the joint work done by me and my colleague. Manual annotation of 200 articles was undertaken collectively to establish a gold set for training and evaluation. My colleague worked on training the model using supervised techniques and I evaluated the QWEN Large Language Model on the same task. Post evaluation the DistilBERT model was chosen to run inference on the collected articles and both article sets were combined to create the relevant article dataset for this study. Subsequent chapters build upon this dataset and represent independent analyses conducted separately by each of us.

To provide training and evaluation data, a sample of 200 articles was drawn from across outlets and labelled manually. Each article was annotated into one of four class-labels:

1. **Conflict-related:** News directly reporting on events in the Israel–Palestine conflict.
2. **Political Statements:** Statements or official positions issued by world leaders, governments, or institutions about the conflict.
3. **Protests and Public Reactions:** Coverage of citizen protests, rallies, or public opinion directly related to the conflict.

4. **Irrelevant:** Articles mentioning Israel or Palestine tangentially (e.g., trade, economics, sport) but not related to the conflict.

For the relevance labels, we mutually decided to keep only conflict related and political statement classes. Since political statements help us understand how outlets frame statements for each side and what imagery is shown in those articles. Inter-annotator agreement was checked through overlap in labelling, and disagreements were resolved through discussion, resulting in a consistent gold set.

4.1 Zero-shot Classification with LLMs

Parallel to section 4.2, I evaluated a large language model on the 4 class classification task using the Qwen-3-235B-Instruct LLM. This model was chosen since it gave the best accuracy to cost trade-off in the class of instruction tuned LLMs.[84]. Enforcing a strict JSON output and batching requests by clubbing 5 articles at a time. The prompt design template is added in the appendix A.3. On the 200 article gold set, this approach achieved an F1-score close to 0.80. In small manual validation checks it was observed that few-shot prompting(adding 5-10 sample headlines with the prompt) produced a few point increase on the F1 score. However, running end to end LLM inference on almost 100,000 was not feasible due to free API limits for the model. Moreover, DistilBERT already achieved a better F1 score in comparison as discussed in section 4.2.

But this experiment showed the efficacy of even low-cost LLMs which can be fine tuned using few-shot prompting to bootstrap additional training data for the supervised model via distillation, i.e. use the LLM results with light manual validation to generate annotated datasets for the DistilBERT to train on and increasing the training dataset can provide better results for supervised classification techniques as well. This retains the LLM's coverage benefits while keeping deployment cheap, fast, and reproducible with the distilled DistilBERT classifier.

4.2 Supervised Classification

Different classification models were trained and evaluated on the gold labelled dataset. Models chosen were the ones that give reliable results on similar 200 labels gold dataset [41]. The models included Logistic Regression, Random Forest, and a Linear SVM, representing strong traditional machine learning baselines, as well as a fine-tuned

DistilBERT transformer model. All models were trained on the 200 manually annotated articles, using an 80/20 train-test split for the traditional baselines. DistilBERT, in contrast, was evaluated using 5-fold cross-validation to provide a more robust estimate of its generalisation ability, given the small dataset size. The results of the comparison are shown in Table 4.1

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.886	0.899	0.869	0.883
Random Forest	0.869	0.848	0.902	0.874
SVM	0.889	0.909	0.865	0.886
DistilBERT	0.900	0.882	0.927	0.903

Table 4.1: Performance comparison of supervised classifiers on the four-class relevance classification task. DistilBERT achieved the highest overall performance.

For DistilBERT, the cross-validation results were averaged over each fold to achieve an overall F1 score close to 0.9. Thus, this model was finalised to be extended to all the articles for filtering the dataset to just have relevant articles.

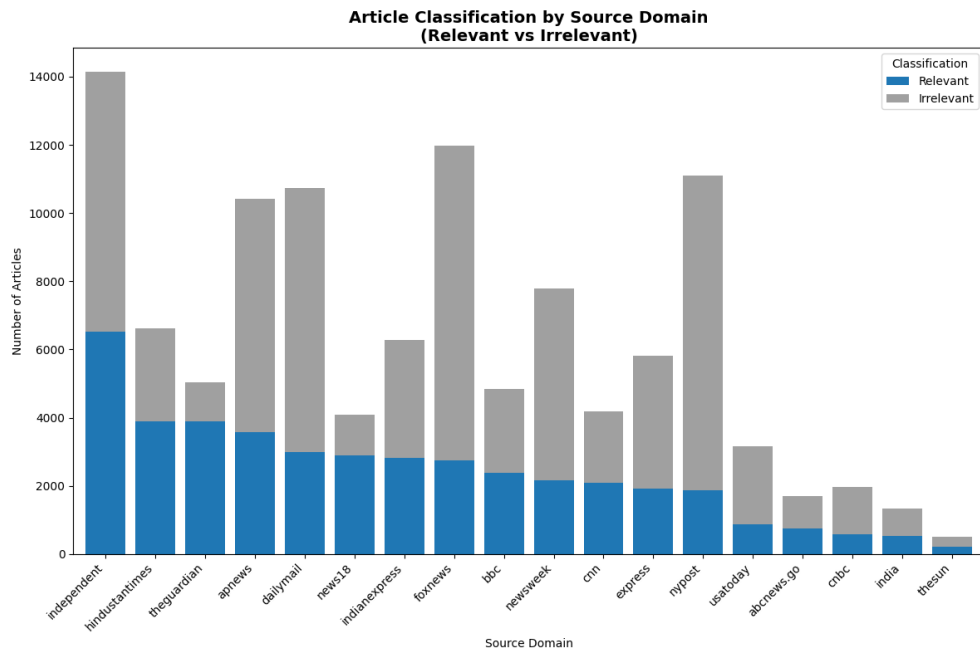


Figure 4.1: Relevant articles filtered for each source

Running the model on all articles, we get a dataset of **42,686** articles with only articles relevant to the conflict. All subsequent chapters operate exclusively on this relevance-filtered set to analyse source-level framing and visual bias over time.

Chapter 5

Methodology

With the cleaned and filtered dataset comprising only of articles relevant to the conflict i.e the conflict related articles and articles reporting on political statements related to the conflict, this chapter describes the methods used to analyse images for bias indicators. The methodology is organised into 4 stages. First the corpus is analysed to extract related lexicons for Israel and Palestine entities. This helped identify cases when terms like IDF, Rafah etc. were used to talk about the sides. Secondly, the images are analysed by mainly using BLIP and CLIP. This was done to convert the image content into text and identify scenes of the images portrayed, i.e destruction, military and victims. Thirdly, these images were tagged to either side, identifying the actor (aggressor) and patient (victim) of the said image using its captions. Finally, the resulting distributions across outlets were compared against event-based baselines using statistical tests to determine whether observed deviations were significant.

Defining Bias: In this study, bias in visual framing is defined as a statistically significant deviation between an outlet’s observed Israel/Palestine split and the expected split from external baselines across the three dimensions - human suffering, armed presence and material damage. Following prior work on media bias and framing in conflict coverage [20, 6], these dimensions were chosen since they are the core elements in visual framing of the conflicts, thus providing a structured lens to study outlet-level differences. We quantify this separately for the three portrayal categories, after scene detection and side attribution, and assess significance using a chi-squared goodness-of-fit test, a positive deviation indicates over-representation of a side relative to baseline, and a negative deviation indicates under-representation. The end-to-end process is summarised in Figure 5.1. We did not adjust for multiple comparisons across outlets/categories; p-values are treated as exploratory, and we emphasise percentage-

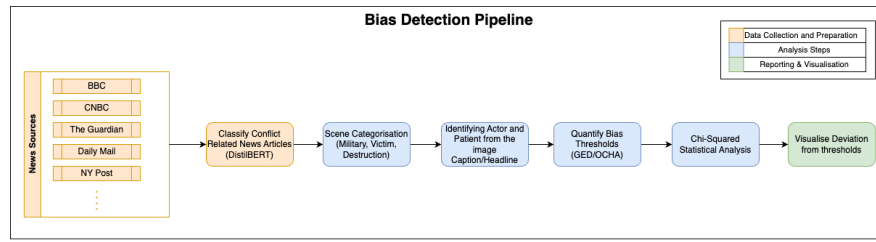


Figure 5.1: Pipeline showing various steps in the study's methodology

point differences as the primary effect size.

5.1 Extracting Terms Specific to Israel and Palestine

The first step of analysing the article images and captions is to find a way to reliably determine whether the images show Israeli entities or Palestinian. Since we are dealing with limited textual indicators like captions provided by the news sources, we need to map other terms like Netanyahu, IDF, Rafah, West Bank to one of the two sides. To achieve this all captions from each source were collated into a single data set and analysed to create entity lexicons for each side. These entity lexicons cover persons, locations, organisations etc., and are agnostic to this dataset and can be re-used in any further study analysing the same domain.

5.1.1 Named Entity Recognition on Image Captions

Following prior bias studies like Raza et al. [59], Named Entity Recognition (NER) was employed to systematically extract entities from both article headlines and captions since transformer based NER has proven to be robust for extracting these entities from small scale texts like captions. It was implemented using the spaCy (v3.8.5) [35] library with its transformer-based English language model "en_core_web_trf" (v3.8). This RoBERTa-based model [43] tags entities as PERSON, ORG, and GPE/LOC (location) tags based on the sentence context. This was run on all the relevant articles to get a named entity vocabulary and then manually classified it to Israel or Palestine. Only terms with frequency over 500 were chosen to weed out noise. Transformer-based NER is well-validated for robust entity extraction across domains, and RoBERTa-initialised pipelines (as used in spaCy's transformer stack) achieve high accuracy on PERSON/ORG/LOC detection, making them suitable for building conflict-specific lexicons at scale [43, 35].

The resulting entity lists were processed to improve consistency. Any variations in spellings were mapped (for ex, Jabaliya and Jabalia) and generic terms like Getty Images etc, were filtered out. For example, settlements and colonies were excluded unless explicitly prefixed as Israeli settlements.

Alternative approaches like unsupervised topic Modelling (LDA) [14] and BERTopic (all-MiniLM-L6-v2) [11] modelling were also tried out, results of which are added to the appendix A.3. However, topics tended to cluster around broad themes such as “conflict” or “politics” rather than specific named actors. Since the goal of this stage was fine-grained side attribution, NER-based extraction was judged more appropriate and adopted in the final pipeline.

The final curated list of lexicons for each side are presented in Table 5.1 (extended list in appendix A.1). This is an illustrative subset of all the lexicons presented for reader’s understanding of entities tagged to both sides. These entity lists were used in later sections for creating regular-expression based heuristic rules and enriching the LLM prompts to tag the identified actors and patients to Israel or Palestine.

Table 5.1: Sample lexicons of entities associated with Israel and Palestine, extracted via NER and curated manually by our module

Side	Category	Entities (examples)
Israel	Core Terms	israel, israeli, israelis, zionist, zionists
	Military & Security	idf, israel defense forces, shin bet, mossad, israeli army, israeli soldiers, israeli troops
	Political Leaders	netanyahu, gantz, lapid, bennett, gallant, smotrich, ben gvir, likud, kneset, israeli government
	Cities & Regions	jerusalem, tel aviv, haifa, beersheba, sderot, golan heights, galilee, negev, kibbutz
Palestine	Core Terms	palestine, palestinian, palestinians, gaza, gaza strip, west bank
	Cities & Regions	rafah, jabalia, deir al-balah, khan younis, ramallah, hebron, bethlehem, nablus, jenin
	Political Orgs	hamas, fatah, plo, pa, pflp, dflp
	Leaders & Factions	abbas, haniyeh, sinwar, meshaal, barghouti, erekat, al-qassam brigades, al-aqsa martyrs brigades

5.2 Image Scene Detection

This stage processes all images to collate the published images that show destruction, military elements and victims. We do this using CLIP since it is proven to be robust against credible benchmarks for image classification[57]. This was achieved through

concise prompt templates for military and victim, and regular expression matching for destruction. Moreover, the missing captions are backfilled using BLIP. Once this method gives us the confidence scores for presence of these scenes in the images confidence thresholds are tuned once on a validation split of 100 manually labelled images, to maximise F1 score, and then applied uniformly across outlets.

To run these experiments, we operate at an image level. The initial article dataset is pivoted to a image level table with `article_id` as foreign key. This ensures all published imagery is analysed, but also means that outlets publishing more images per article may contribute disproportionately. This is the input dataset that would be used throughout this section.

5.2.1 Caption backfilling with BLIP

To handle cases where the caption was missing from the images published by the news outlets, BLIP (Bootstrapping Language-Image Pretraining) was used to generate captions describing the content of the image in depth. Prior studies have shown that BLIP produces semantically consistent captions [40]. This method was implemented for all images and used only for detecting the scene since BLIP cannot identify actors, etc. It was only used where captions were missing so as to give more priority to publisher-provided captions. This ensured no image was excluded from our analysis. BLIP captions were never used for side attribution.

5.2.2 CLIP based Scene Classification (Victim & Military)

To detect military or victim indicators from images, CLIP (Contrastive Language-Image Pretraining) with the public checkpoint `openai/clip-vit-base-patch32` (ViT-B/32, 224×224 input) was used in zero-shot mode (no fine-tuning) which has shown good results for zero-shot QnA in prior studies [57]. The images were resized as per specification and exact prompts used are listed in appendix A.8. Since the same image can be tagged to both classes, the detectors are run sequentially first for military then for victim identification. Each detector uses prompt templates for positive and negative cases for e.g. “a photo of armed personnel military or militants or military vehicles” (military positive) or “a photo showing wounded people.” (victim positive); negative prompts are logged for sanity checks but not used in the decision rule.

On inference, this gives us positive scores for each classification, which are essentially the cosine similarity scores between the image and the prompt. These scores are

then tuned on a manually labelled dataset of $N=100$, to maximise the F1-score. The threshold is increased in increments of 0.1 and local refinements later on 0.01 near the optimum and the F1 score is calculated at each score value. Validation images were sampled across outlets and labelled manually by the author. On completing this exercise, the threshold for military was fixed at **0.25** and for victims is set at **0.278**. These thresholds are then applied across all outlets to collate image subsets which are tagged to military and victims, which come up to be just under 10,000 images per scene. All experiments were implemented in PyTorch (HuggingFace Transformers), and both cosine similarity scores and binary predictions were stored per image to be used in subsequent bias quantification.

5.2.3 Regex-based Scene Classification (Destruction)

To identify the images depicting destruction, the captions (along with BLIP backfilled captions) were scanned for lexicons related to destruction like rubble, ruin, devastate, demolish, crater, and bomb. These *regular expression* patterns were applied after converting the captions to lowercase. This approach was used since destruction images were generally seen to be explicit in defining the destruction using textual captions as compared to the visual method used with victim and military classification.

To validate these methods, again a set of 100 sampled images across outlets was manually labelled by the author. The regex achieved high coverage with precision, recall and F1 score as shown in Table 5.2.

5.3 Evaluation of Scene Detection

For all three classifications (military, victim, destruction) the author manually labelled 100 randomly sampled images across outlets. In each sample, the images were labelled to show or not show military, victims or destruction. For regex, this was straightforward and resulted in an F1 score of 0.88 on the validation set. For CLIP-based methods, multiple thresholds were tried by incrementing the similarity score between the image and the prompt. Finally, the F1 scores above 0.9 were achieved for both at thresholds of 0.25 and 0.278, respectively. The final class-wise precision and recall scores are added to the appendix for a deeper dive A.6.

Table 5.2: Final selected decision rules and validation performance by category.

Category	Best threshold	Precision	Recall	F1-Score
Military	0.25	0.92	0.885	0.902
Victim (Casualty)	0.278	0.957	0.936	0.946
Destruction	Regex Based	0.878	0.89	0.884

These high accuracy scores in Table 5.2 instill confidence that the methods to classify the presence of military, victim and destruction elements in the images are sufficient for bias quantification.

5.4 Attributing Images to Conflict Sides (caption-based)

The next logical step in this study was to evaluate the direction of violence shown in the military, destruction and victim images collected so far by analysing the accompanying caption texts. This was one of the more challenging parts in the pipeline given the inherent ambiguity in the captions(e.g. "scene of attack" or "photo captured from across the border") and importance of the attribution to analyse the bias in framing tied back to the research questions.

5.4.1 Early Attempts: Rule-based and Structured NLP Approaches

Initially, experiments were conducted to use the NER lexicons generated in section 5.1.1 in combination with the two approaches mentioned below.

1. **Semantic Role Labelling:** Using AllenNLP's SRL [27] model to extract verbs, agents and patients from the captions for each category(military, victim, destruction). This would be followed by extracting violent verbs and tagging the agent as Israeli or Palestinian using the NER entity list.
2. **DistilBERT SQuAD via HuggingFace:** Using the HuggingFace, distilbert-based-distilled-squad Question-Answer model on the relevant captions from each of the retrieved categories [58]. For each caption, fixed prompts like "Who caused the destruction?" & "What is the location of destruction?" were designed. Similar prompts are designed for the military and the victim as well. Then these are mapped to the NER entities to tag Israel or Palestine as aggressor or victim of the violence.

To gauge performance, we manually labelled 100 randomly sampled images per category (destruction, military, victim) to create gold-standard sets. When the structured baselines (NER+SRL / NER+QA) were applied to destruction captions, performance deteriorated due to the brevity and implicit phrasing typical of captions; models fine-tuned for QA on longer Wikipedia-style passages struggled to extract decisive spans. In the QA+heuristics variant, Israel achieved very low precision (0.32) despite perfect recall (1.00), indicating systematic over-assignment to Israel (many false positives). Palestine performed better but still modestly ($P=0.49$, $R=0.72$). Overall, macro $F1=0.53$ on destruction captions, confirming that even the hybrid variants remained fragile and insufficient for reliable side attribution at scale.

Class	Precision	Recall	F1-score	Support
Israel	0.32	1.00	0.48	32
Palestine	0.49	0.72	0.58	68
Overall F1 (micro / macro): 0.54 / 0.53				N=100

Table 5.3: Destruction attribution-QA baseline on 100 captions (None/Unclear excluded).

5.4.2 Method Used: LLM-Based Side Attribution

To address the nuanced context in small texts like captions and headlines, side attribution was implemented with an LLM, GPT-3.5 Turbo (OpenAI API) was used since it is cost-effective, works well with batched JSON requests and understand implicit references well as shown by cross domain studies like Loukas et al.[44] The exact prompts used can be seen in the appendix A.8. In general, the model was run sequentially on each of the three tasks to assign the agent label as Israel, Palestine or Other/None. To ensure consistent outputs, the prompt was engineered to return a strict JSON object for each caption, identifying the aggressor for military and the victim in the victim and destruction categories.

The side attribution is kept limited to the caption to prevent source leakage and keep the classification solely based on the current image. Inference is deterministic by keeping temperature=0. Also, the model is allowed to return Israel, Palestine and neutral labels so as to prevent mislabeling unclear captions like captions talking about Israel’s war with Lebanon. Captions are processed in batches of 20, and then the result is post-processed as a JSON, ensuring consistency of the results.

Each label is joined with the image set gathered in section 5.2 to get the location for destruction, victim’s country for victims and military affiliation for military images.

Now **evaluation** of these results against the earlier labelled gold-standard data sets show massive improvements (see table 5.4). It is important to note that destruction classification precision is still below 0.9 driven primarily due to the destruction captions sometimes not mentioning the victim only the aggressor, for e.g. Israeli airstrikes kill twelve. Also, elliptical phrasing may inflate false positives for the Israel class.

Task	<i>N</i>	Precision (macro)	Recall (macro)	F1 (macro)
Destruction	86	0.860	0.980	0.910
Military	100	0.991	0.989	0.990
Victim	94	0.987	0.991	0.989

Table 5.4: GPT-3.5 Turbo side attribution evaluation by portrayal category.

Overall, the GPT-3.5 Turbo approach provides reliable, deterministic, caption-led attribution and replaces the brittle rule-based variants as the core method; unclear items are retained but excluded from comparative analyses. Any broader concerns about model bias and interpretability are addressed in the chapter-end methodology limitations. The resulting side labels, joined to the scene subsets (destruction, victim, military), enable outlet-level observed Israel/Palestine proportions that we now test against externally grounded baselines.

5.5 Bias Quantification

With the labelled datasets ready for all 3 scenarios, we can analyse them to see the difference in framing such event imagery across the 17 outlets. However, to effectively analyse if these outlets over or under-represent any events, real on-ground statistics are needed for military deployments, victims (casualties) and destruction caused. Methods for getting the baselines for each category is specified below:

1. **Victims:** UN OCHA (Office for the Coordination of Humanitarian Affairs) casualties [77] (see A.10) derived from the OCHA Casualties dashboard by summing the injuries and fatalities for the period of the study, i.e., Oct 2023–May 2025. This gives the threshold values: **96.9%** Palestinian vs **3.1%** Israeli.
2. **Destruction:** UCDP GED [78] (see A.10), Geo-referenced events dataset used to get the location of destruction based on the event coordinates lying in Israeli or Palestinian territory. This gives **92.3%** Palestinian vs **7.7%** Israeli.

3. **Military:** No neutral deployment baseline since military deployments are not released to the public. We use the GED thresholds as an anchored proxy since Gaza-side destruction is indicative of Israeli action (consistent with conflict asymmetry) and, as a sensitivity check, the cross-outlet averages are also used as a more conservative threshold. We state the anchor per result and ensure that effect directions are robust under both.

Composite bias score. To summarise results across dimensions, the cross-outlet average can't be used since the average would itself be skewed based on the outlets, so we use a representative scale,

$$\text{Score} = P^V + P^D - P^M, \quad (5.1)$$

where P^V , P^D , and P^M are the Palestinian shares (percent) in *victims*, *destruction*, and *military*. Higher scores indicate more Palestinian harm (victims/destruction) and less Palestinian military; lower scores the reverse (Israeli-leaning). The score is descriptive; inference relies on per-portrayal two-sided proportion tests against event baselines (OCHA, GED). Low-coverage cells are reported but not over-interpreted.

5.5.1 Statistical Significance Testing

In each task and for each outlet, we compute the observed counts of Israel- and Palestine-labelled items over the full study period. By getting the count of images (N) in each label, the chi-squared and p-values are calculated for the **chi-square goodness-of-fit test** [51] to calculate the deviation from the actual event baselines and confirm whether it is statistically significant. For our test, the null hypothesis (H0) is that the outlet's side distribution matches the baseline proportions. The alternative hypothesis (H1): the outlet's side distribution differs from the baseline. The deviation is said to be significant if the p-value is less than 0.05. Tests are only interpreted when both expected counts are adequate (rule-of-thumb N greater than 50); otherwise, the comparison is flagged as an insufficient sample. This completes the operationalisation of bias for our study: bias is defined as a statistically testable deviation between an outlet's caption-led side distribution and an externally grounded, full-period baseline (UN OCHA for victim; GED for destruction; actor-coded GED or a conservative pooled average for military). The resulting outlet-level statistics are presented in the Results chapter and form the basis for the interpretation that follows.

5.6 Key Limitations

While the pipeline is comprehensive and evaluated robustly, some considerations must be noted. The usage of pretrained large language models like GPT 3.5 Turbo (for tagging sides) and CLIP (for identifying the scene of the image) introduces the possibility of **AI bias** that depends on the data the models are trained on. Since these models perform better than the other heuristic approaches discussed, we minimise these risks by keeping side attribution completely caption-led so it does not club similar outlet articles together, using a constrained JSON schema for output parsing and running the model with temperature=0 to make the outputs reproducible. For CLIP, the model was validated against a manually annotated validation set to tune the CLIP thresholds to maximise F1 score and then frozen for all outlets. These choices reduce model bias, but do not eliminate it; provider/backend updates can still shift outputs slightly.

The bias baselines were sourced from credible academic and neutral organisations (like the United Nations) to ensure thoroughness. However, it is acknowledged that these **baselines could vary from the ground truth** as they can contain reporting lags or incomplete coverage. To check for robustness, the outlets' Israel-Palestine splits were also compared against cross-source pooled average distributions and the higher biased outlets for each category (military, destruction, victim portrayal) continued to show clear and consistent deviations, indicating that the results are not an artefact of baseline choice but reflect genuine framing differences across outlets as summarised in the results later.

Finally, the evaluation of each of the methods used was constrained to a separate **validation set** to ensure robustness however, these sets were **limited to 100 randomly sampled images** per category. Random sampling allowed for fair testing of each component separately, but may introduce risks of single annotator bias and the possibility that results may vary slightly on different random draws. These risks were minimised by ensuring samples drawn were completely random and applied consistent labelling guidelines focusing only on the Israel-Palestine conflict and providing room for uncertain/None labels for ambiguous cases. Taken together, these steps minimise the risks of biases, making the reported results reproducible within the stated limits.

Chapter 6

Results and Discussion

This chapter presents the findings of visual framing bias as discussed in the previous chapter across 3 dimensions: **military portrayal**, **victim portrayal** and **destruction portrayal** by images and captions across Israeli and Palestinian actors. The distribution of images across actors analyses the direction of violence portrayed by the imagery and finally uses the chi-squared goodness of fit test to see if the distribution varies from actual event baselines/thresholds significantly.

6.1 Military Portrayal

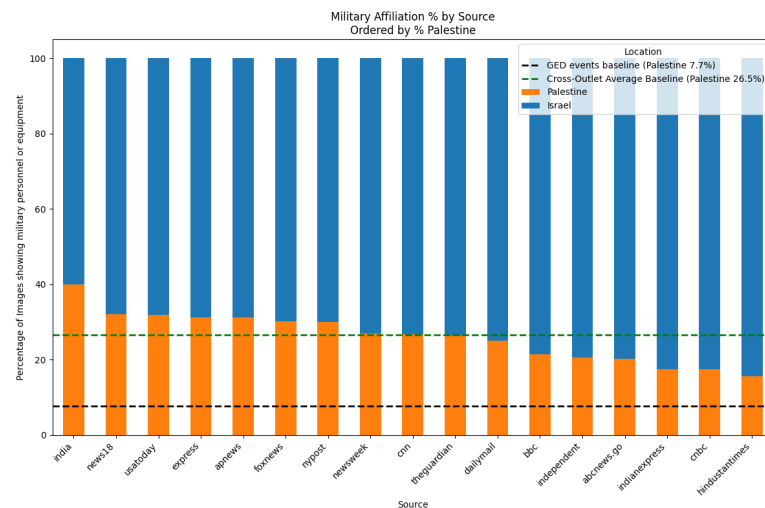


Figure 6.1: Stacked percentage bar chart showing Palestine (orange) and Israel (blue) military portrayal across news sources, with dashed line indicating GED events baseline as a proxy (7.7% Palestine) and overall distribution average (26.5% Palestinian).

Figure 6.1 shows the percentage of Israeli (blue) vs Palestinian (orange) military images published across the 17 outlets. For comparison, the event baselines from the GED (Georeferenced Event Dataset) [78] event data of 7.7% Palestinian conflicts and dataset average (26.5% Palestinian military imagery) are added to contextualise the deviations from on-ground events. All outlets substantially deviate from the GED Israel-Palestine conflict event distribution baseline, with the lowest, Hindustan Times, still showing double the expected proportion i.e 15.5% Palestinian military portrayal. The significance test results confirm that the statistical significance of these deviations using a **chi-squared goodness of fit** test against the event baseline, helping establish that these deviations are significant and not random chance. All outlets show a p-value less than 0.05, reinforcing that the observed disparities are systematic rather than coincidental.

Relative to the cross-outlet average baseline (26.5%), the outlets cluster into 2 distinct groups. Nine above the baseline and eight below. The largest deviation towards over-representing the Palestinian military is shown by India and News 18. Similarly, on the opposite spectrum, Hindustan Times, CNBC and Indian Express show the most negative deviation from the average baseline, indicating that these sources show more Israeli military images as compared to other sources. Newsweek, CNN, The Guardian and Daily Mail lie in the middle, closer to the cross-source average but still 3-4x more than the GED baseline.

Our analysis shows that on both sides of the spectrum, Indian sources show extremes with India, News 18 over-representing the Palestinian military, whereas Hindustan Times and Indian Express seem to be portraying the Israeli military more in their published images. Similarly, US outlets like Fox News, NY Post sit above the baseline to a significant degree, showing 3-5% more Palestinian military than the rest, showing a trend of over-representing Palestinian military. The UK sources (The Guardian, BBC, Daily Mail) sit closer to the average, showing minor deviation from the average baseline. However, Express is the exception here, being the only UK source lying above the baseline.

While these findings indicate a consistent over-representation of Palestinian-affiliated military imagery across outlets, the potential role of editorial selection norms and model classification biases is acknowledged. These nuances are explored in greater depth in the final discussion section 6.4.

Table 6.1: Sample images for military portrayal classification.

Source	Image	Caption	Military Affiliation
foxnews		Terrorists from Hamas during an anti-Israel military march in Gaza City.	Palestine
dailymail		Hamas claimed Israel had violated the ceasefire agreement that was agreed last month	Palestine
apnews		Hamas fighters attend the funeral procession of Samer al-Haj, a Hamas official who was killed by an Israeli drone strike, at Ein el-Hilweh Palestinian refugee camp, in the southern port city of Sidon, Aug. 10, 2024.	Palestine
apnews		Israeli soldiers move on armored personnel carriers near the Israeli-Gaza border as smoke rises to the sky in the Gaza Strip, seen from southern Israel, Sunday, Jan. 21, 2024. (AP Photo/Leo Correa)	Israel
foxnews		Israeli military vehicles are seen traveling in the city of Kiryat Shmona, in northern Israel	Israel
dailymail		Israeli troops patrol in armoured personnel carriers in Israel	Israel

6.2 Victim Portrayal

Figure 6.2 shows the distribution of Palestinian vs Israeli victim imagery shown across the 17 analysed outlets. The baselines they are compared against are the UN OCHA [77] casualty dashboard, which indicates the injured and fatality distribution to account for 96.9% Palestinian casualties, and a more conservative threshold of the average distribution across all outlets, which comes out to be 77.1%. Relative to the OCHA victim baseline, all sources publish lower Palestinian victim shares and Israeli victims are over-represented as compared to the on-ground proportions. Chi-squared goodness of fit test against the OCHA baseline shows that the deviations are significant for all outlets ($p < 0.05$). Establishing that these deviations are statistically significant.

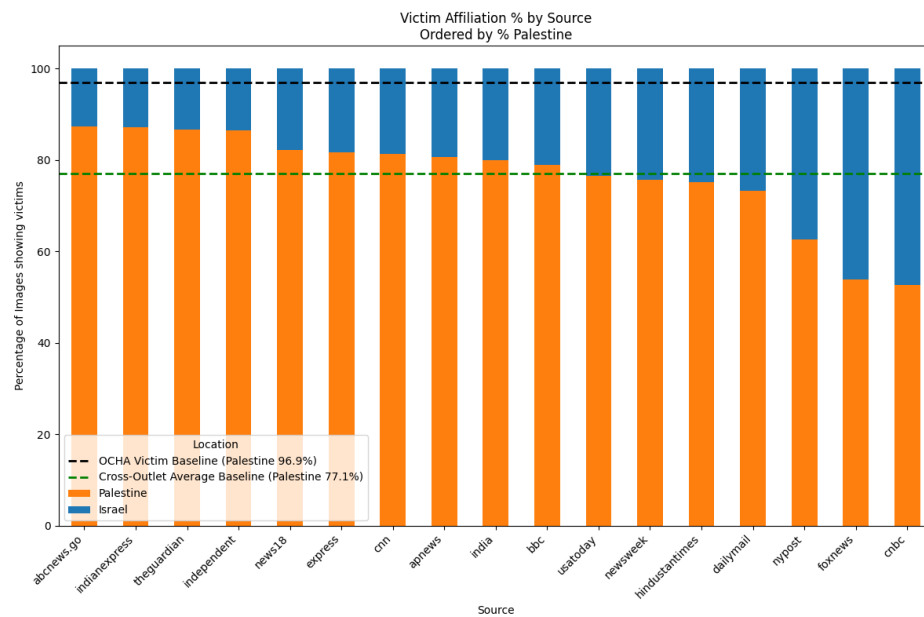


Figure 6.2: Stacked percentage bar chart showing Palestine (orange) and Israel (blue) victim portrayal across news sources, with dashed line indicating UN OCHA casualty count baseline (96.9% Palestinian casualties) and overall distribution average (77.1% Palestine).

Relative to the cross-outlet average (77.1% Palestinian victims), the outlets split into 2 groups, showing 10 outlets above the baseline and the rest below, over-representing Israeli victims. The largest positive deviations are shown by ABC News, Indian Express and The Guardian, showing Palestinian victims closer to the ground truth. It was observed that the degree of deviation on the other end was much larger with Fox News, NY Post and Daily Mail over-representing Israeli victims 5-20% more as compared to the sources in the middle. These sources in the middle show deviation less than 2% of the cross-outlet average, and this cluster comprises sources like USA Today, BBC, India, News Week and Hindustan Times.

Clustering by region, UK outlets (The Guardian, BBC) tend to sit closer to the baseline or slightly over-representing Israeli victims (Daily Mail). US outlets are bimodal; Fox News and, NY Post show the most over-representation of Israeli victims; however, it is balanced by AP News, CNN, and USA Today, which show fewer Israeli victims than the average. For victim portrayal, Indian sources are mostly above the baseline and closer to the ground truth casualties, as shown by the OCHA baseline.

The sample tagging in Table 6.2 from outlets with the highest deviations illustrates how visual framing through choice of subjects, context, and composition can amplify

or downplay victimisation for each side.

Table 6.2: Sample images for victim portrayal classification.

Source	Image	Caption	Victim Affiliation
dailymail		Israeli soldiers embrace next to photos of people killed and taken captive by Hamas militants during their violent rampage through the Nova music festival in southern Israel	Israel
dailymail		Two-month-old Muhammad Al-Khayyat baby dies in the hospital despite all the efforts as a result of the Israeli attacks in Rafah, Gaza on October 13, 2023	Palestine
foxnews		Mourners attend the funeral of the Kotz family in Gan Yavne, Israel, on Tuesday, Oct. 17. The Israeli family of five was killed by Hamas terrorists on Oct. 7.	Israel
nypost		A man standing next to burnt cars points at the damage in his house, a day after an attack by Jewish settlers on the village of Jit near Nablus in the occupied West Bank that left a 23-year-old man dead	Palestine
nypost		Soldiers react next to the coffin of Israeli soldier Sergeant Omer Ginzburg, who was killed amid the ongoing conflict in Gaza between Israel and Hamas during his burial in Kiryat Tiv'on, Israel, August 13, 2024.	Israel
foxnews		Mourners attend the funeral of Palestinian Hani Al-Jaafarwi, the director of ambulance and emergency services at the ministry of health, who was killed in an Israeli strike on Al-Daraj clinic in Gaza City on June 24, 2024.	Palestine

6.3 Destruction Portrayal

On comparing the results of the destruction imagery and tagging it to the location where the destruction occurred, Figure 6.3 shows the distribution of Israel vs Gaza destruction images across the 17 analysed news sources. The baseline used here is the percentage derived from the University of Uppsala's (Sweden) GED [78] conflict event dataset, which comes out to be 92.3% Palestinian destruction, as per real conflict events. For additional context, the conservative, cross-outlet average baseline is also added for destruction, which stands at 76.1% Palestinian destruction. Relative to the GED baseline, all outlets show significant deviation from the GED baseline, under-

representing Palestinian destruction. The chi-squared test against the GED baseline is significant for all outlets except CNBC since the image count for CNBC where the destruction-image count is small.

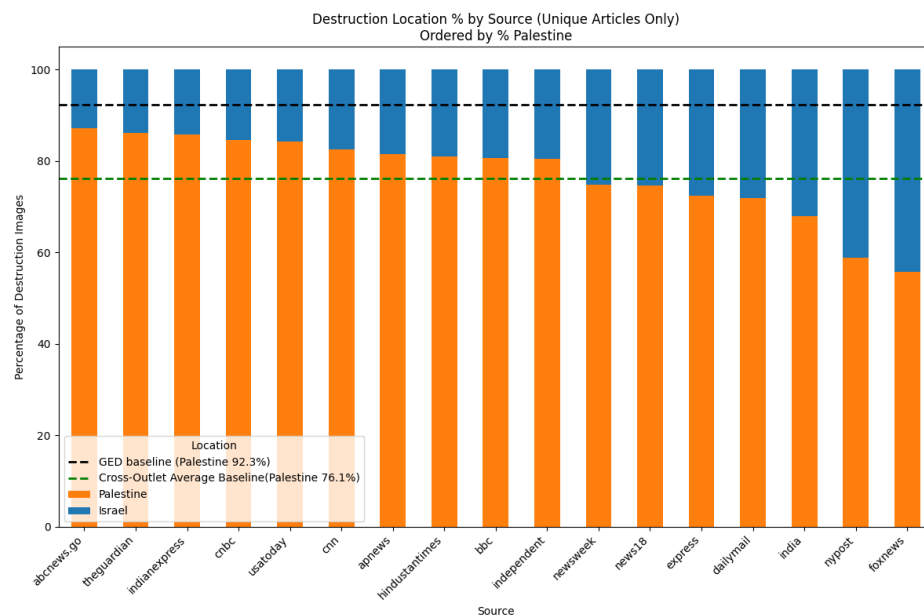


Figure 6.3: Stacked percentage bar chart showing Palestine (orange) and Israel (blue) destruction portrayal across news sources, with dashed line indicating GED events baseline (92.3% Palestine) and overall distribution average (green) – 76.1% Palestine destruction.

Relative to the cross-outlet average, outlets are split into 2 groups: 10 outlets inching closer to the GED baseline and 7 outlets that under-represent Palestinian destruction. A notable observation is that the degree of deviation for the under-representation of Palestinian destruction is 2x that of deviation showing more Israeli destruction on the other end. Again, the largest negative deviations are exhibited by Fox News and the NY Post. Indian Express, ABC News and The Guardian show more Palestinian destruction as compared to the average source. Near baseline cluster comprises Newsweek and News 18.

By region, US outlets are skewed to over-represent Israeli destruction due to the large deviation shown by Fox News and the NY Post. CNN and USA Today show Palestinian destruction more than others, but to a small degree. UK Outlets tend to cluster closer to the baseline, with the exceptions being the Daily Mail and Express, which under-represent Palestinian destruction. Indian outlets are mixed: Indian Express and Hindustan Times above baseline, India below, and News18 near baseline. Example

frames from the top outlets are shown in Table 6.3 to illustrate how these images were tagged using the captions.

Table 6.3: Sample images for destruction portrayal classification.

Source	Image	Caption	Destruction Location
dailymail		Israeli soldiers walk past a house destroyed in the attack by Hamas terrorists	Israel
dailymail		A Palestinian woman walks among the rubble in the residential area known as Juhor ad-Dik in the Gaza Strip on Tuesday	Palestine
nypost		Rubble in Kibbutz Kfar Aza, which was attacked by Hamas on Oct. 7.	Israel
nypost		Khan Younis, southern Gaza's largest city, has been reduced to rubble following months of intense fighting between Hamas and the IDF.	Palestine
foxnews		Destroyed cars and personal effects are shown scattered around the Supernova Music Festival site, where hundreds were killed and dozens taken by Hamas militants near the border with Gaza in Kibbutz Re'im, Israel. (Alexi J. Rosenfeld/Getty Images)	Israel
foxnews		GAZA CITY, GAZA - OCTOBER 18: A boy tries to collect usable belongings amid wreckage of vehicles after Al-Ahli Baptist Hospital was hit in Gaza City, Gaza	Palestine

6.4 Discussion

This section discusses the outlet-level portrayals and how they shape the visual framing bias across the military, victim and destruction portrayals for Israel and Palestine. The goal is to analyse bias patterns when these portrayal values are compared to the actual event baselines, and the significance of deviations is analysed through statistical tests. Moreover, we compare the visual framing summarising results across all three dimensions to analyse how these outlets compare to each other.

When compared to credible event baselines like GED [78] and OCHA [77], across

Table 6.4: Bias score across outlets based on Palestinian shares in victims (P^V), destruction (P^D), and military (P^M). $Score = P^V + P^D - P^M$, where higher values indicate relatively more images of Palestinian victims and destruction and fewer images of Palestinian military actors (leaning Palestinian), while lower values indicate the opposite (leaning Israeli). All percentages represent the share of images attributed to Palestinians in each portrayal category. Outlets are sorted by score from lowest (Israeli-leaning) to highest (Palestinian-leaning).

Outlet	Victims	Destruction	Military	Score
	P^V (%)	P^D (%)	P^M (%)	
Fox News	53.8	55.8	30.2	79.4
NY Post	62.7	58.8	30.0	91.5
India	80.0	68.0	40.0	108.0
CNBC	52.6	84.6	17.4	119.9
Daily Mail	73.3	71.9	25.0	120.1
Express	81.6	72.4	31.3	122.8
Newsweek	75.7	74.8	26.9	123.6
News 18	82.2	74.6	32.1	124.7
USA Today	76.5	84.3	31.8	128.9
AP News	80.7	81.5	31.2	131.0
CNN	81.3	82.5	26.7	137.1
BBC	79.0	80.7	21.4	138.3
Hindustan Times	75.2	81.0	15.5	140.6
Independent	86.4	80.5	20.6	146.4
The Guardian	86.7	86.1	26.4	146.5
ABC News	87.4	87.2	20.2	154.5
Indian Express	87.2	85.8	17.5	155.4

outlets, systematic deviations are observed from the baselines. The distribution graphs in Figures 6.1, 6.2, 6.3 show that all outlets tend to over-represent Palestinian military elements and under-represent Palestinian victims and destruction in varying degrees. These results are verified using chi-squared goodness of fit tests, showing that these deviations are statistically significant (p-value less than 0.05).

To summarise the results across the three dimensions, we use the composite bias score scale discussed in Section 5.5 (see equation 5.1). Since the score is based on Palestinian harm and agency, a higher score indicates comparatively more Palestinian harm (victims/destruction) and less Palestinian agency (military), and lower scores indicate the opposite. Thus, lower-scoring outlets tend to align more with Israeli interests. This helps us gauge the degree of bias portrayed by combining results across all three dimensions. The results of scoring all outlets with the score are shown in Table 6.4 sorted in ascending order from most Israel-supported visual framing to least.

Analysing the cross-dimensional scores for each outlet from the Table 6.4, three patterns stand out. First, outlets occupy a wide band of score values, i.e 79-155 on a scale that ranges from -100 to 200. This implies consistent differences in framing among outlets rather than random fluctuation. Second, outlets with the lowest scores, Fox News, NY Post, India, and Daily Mail show that these sources consistently show fewer Palestinian harm (destruction, victims) and more Palestinian agency in violence (military actors). Finally, a high score group comprising Indian Express, ABC News, The Independent and The Guardian shows a more balanced portrayal that is closer to the on-the-ground baselines like GED and OCHA. We want to emphasise here that these scores do not infer intent but show a descriptive summary of observed content selection.

A notable pattern here is that the scores in the table also form regional clusters to an extent. The US outlets are bimodal, with Fox News and NY Post showing the lowest scores and thus the highest over-representation of Israeli harm and under-representation of Israel's armed involvement, whereas USA Today, AP News, CNN and ABC News occupy mid to upper regions in this distribution. The UK outlets are generally mid to high range, showing a balanced portrayal when compared to all analysed outlets, with BBC, The Guardian, and The Independent being closer to the middle, with the exception of Daily Mail, which tends to under-represent Palestinian harm. Indian outlets span the spectrum, with India and News 18 being relatively low and the Indian Express having one of the highest scores. These patterns suggest region-level tendencies with substantial within-region heterogeneity

Diving deeper into other plausible reasons for this observed disparity in reporting could be due to the news outlets' favouring conflict and action, which makes showing armed Palestinian forces a better avenue for them, rather than slow-moving Israeli military images, as backed by several seminal studies in journalism [26, 34]. Moreover, it can also depend on the availability of images. It can be the case that Israeli army pictures are mostly captured by locals or government sources that often show stationary images or images showing regular patrolling (inaction), whereas Hamas-related images are usually captured in an aggravated setting by wire services on the ground nearer to the conflict as backed by studies on embedded reporting like Schwalbe and Pfau [63, 53]. Taken together, these structural factors like news values, access constraints, and sourcing pipelines help explain why some streams of imagery may emphasise Palestinian attributed agency (armed actors) and Israeli-attributed harm (victims/destruction), a substantial literature on visual framing shows such choices can shape audience perceptions and judgements [23, 24, 64, 29].

Several steps were taken to ensure inference quality is maintained. The comparison was made against external baselines sourced from well-recognised neutral bodies. A more conservative average baseline was also used to contextualise the over/under-representation. It was ensured that all outlets were compared against the same time frame, and low-coverage outlets like "The Sun" were excluded from the analysis. Also, unequal sample sizes for images were handled by the analysis being done on portrayal proportions combined with the custom bias scale (Equation 5.1) though the equal weighting of portrayal dimensions in the composite score remains a simplification that may affect rankings. Cases where the no. of images was less than 100 were not analysed harshly. Some limitations of the methodology may be the reliance on GPT3.5 for side attribution, whose training data may have model bias; however, manual validation and precision, recall evaluation show high indicate high accuracy, though the equal weighting of dimensions in the score is a simplification.

Revisiting the research questions:

RQ1 (quantification): For visual bias quantification, images and their accompanying captions were analysed to extract the agent and patient relationship from the images across three portrayals, i.e destruction, victims and military to identify Palestine and Israel as the aggressor or the victim. Each outlet's percentage share was calculated and compared against event-based baselines; deviations from the cross-outlet average were also reported for descriptive comparison, and results were summarised with the composite bias score. This helped uncover patterns in which sources over-represent Israel as the aggressor or Palestine as a victim.

RQ2 (how bias manifests in visual contexts): By analysing these image analysis percentage shares and benchmarking them against event baselines, systematic trends emerged showing over-representation of Israeli harm and under-representation of Israeli agency across sources. Moreover, on the basis of the composite bias score, region-wise source clusters emerged, showing consistent results across the three portrayals - victim, military and destruction.

RQ3 (comparison to event baselines and significance): Running the significance tests against the event-based baselines, all deviations exhibited a p-value less than 0.05 except CNBC (due to low image coverage for destruction), thus showing these differences are systematic and not by chance.

In conclusion, the analysis reveals a consistent outlet-level structure in visual portrayals with significant deviation from the OCHA or GED event baselines and stable source clusters formed across portrayals, showing similarities in publishing policy.

Chapter 7

Conclusion and Future Work

This study aimed to analyse bias in reporting in times of conflict through the lens of visual bias framing during the ongoing 2023-2025 Israel and Palestine conflict. A multi-source, multimodal domain-specific dataset of more than 100,000 articles spanning 17+ source websites was assembled to create an analysis corpus with associated images and caption metadata. An analysis pipeline was built over this dataset to compare framing of victims, destruction and military actors for each side in the conflict, along with systematic evaluation techniques to ensure the pipeline is robust. To our knowledge, this is one of the largest multi-modal datasets built solely for the Israel-Palestine conflict, reporting other data sets either cut across domains or only deal with text [19, 76, 12]. This large-scale corpus enabled us to analyse outlet-level bias in visual framing and run statistical tests to ground our analysis results.

With the research methodologies, this work contributes a scraping strategy to utilise the same data collection pipeline to collect similar datasets across various other domains as well. The image analysis pipeline demonstrates how visual framing can be tested and analysed by using on-the-ground event baselines and caption metadata. Combining state-of-the-art techniques like **BLIP** [40], **CLIP** [57] and **LLMs** [84, 44] to analyse images at scale and identify the aggressor and victims portrayed in the imagery published. Combined with baseline-aware significance testing and outlet-wise error checks, this demonstrates how large-scale, automated visual analysis can be conducted with guardrails appropriate for conflict reporting.

On a high level, it was found that the outlet portrayal diverged in number from the baseline realities. When comparing between peer sources using the cross-outlet average, and the composite bias score based on all three dimensions, some outlets tend to over-represent Israeli military actors by a significant degree, while others stay close to

or under the average baseline. Similar inferences were drawn for victim and destruction portrayal as well. Thus, showing that the same outlet can deviate on one dimension but align with the peer average on the other, emphasising the need for analysing dimensional bias patterns.

The design of the overall pipeline is documented well and is modular. The pipeline is generalisable to other use cases across domains which rely on image analysis and deriving results based on the image metadata (captions) provided. Scene classification and side attribution can be generalised to other use cases by modifying the prompts in place and tuning the acceptance thresholds. Event-based baselines can be swapped to generalise this pipeline to analyse media bias across other conflicts or modify this pipeline to other large-scale image analysis work-streams without re-architecting.

7.1 Future Work

Future extension to the work can prioritise **richer baselines** by combining multiple on-ground baseline sources to analyse deviations per outlet, addressing uncertainties that arise due to the overall pattern of deviation from ground truth. **Finer visual semantics** can be analysed to give a deeper analysis across dimensions, like analysing how different outlets report on *airstrike aftermath* or *medical evacuation*. **Temporal analysis** can be included as another dimension to show how the language evolves for different sources over time, for example, by tracing nearest-neighbour terms for “Israel” and “Palestine” around key real-world events, in line with small-scale lexical-drift studies[69] using word embeddings. Another refinement can be to include **multi-annotator** human verification for the labelled datasets and increase the labelled datasets coverage to make the evaluation strategies even more robust. Finally, the current analysis focuses on English language articles, this **analysis can be extended to Arabic or Hebrew** news outlets to better analyse bias in regional media and also include **cross-platform** analysis from broadcast and video allowing measurement of **video bias indicators** like spoken cues or interruptions alongside textual and visual framing, yielding a more comprehensive view of bias.

This study’s contribution, in sum, is an adaptable blueprint to analyse visual bias in conflict reporting. By combining large-scale data collection, baseline-aware analysis and robust evaluation techniques, this study aims to show how visual media bias can be studied and monitored across various dimensions. It also hints at avenues of adapting this same pipeline to analyse bias across different conflicts or reporting themes.

Bibliography

- [1] Hundreds of residents in the gaza strip have fled their homes to move away from the border with israel., 2023. URL: <https://www.aljazeera.com/gallery/2023/10/7/photos-israel-army-launches-air-attacks-on-gaza>.
- [2] War on gaza: How language used by media outlets downplays palestinian suffering, Nadda Osman. URL: <https://www.middleeasteye.net/news/war-gaza-how-media-language-manipulated-justify-killing-palestinians-and-u>
- [3] ABC News. Israel-hamas war 1 year: At least 41,534 killed in gaza, 1,200 in israel since hamas attack, October 2024. Accessed: 2025-08-21. URL: <https://abcnews.go.com/International/israel-hamas-war-death-toll-1-year/story?id=114458943#:~:text=Israel%2DHamis%20war%201%20year,attack%20in%20Israel%20on%20Oct>.
- [4] E. S. Agung, A. P. Rifai, and T. Wijayanto. Image-based facial emotion recognition using convolutional neural network on emognition dataset. *Scientific Reports*, 14:14429, 2024. doi:10.1038/s41598-024-65276-x.
- [5] Al Jazeera. One year of israel’s war on gaza, by the numbers, October 2024. Accessed: 2025-08-21. URL: <https://www.aljazeera.com/news/longform/2024/10/8/one-year-of-israels-war-on-gaza-by-the-numbers>.
- [6] Wael F. Al-Sarraj and Heba M. Lubbad. Bias detection of palestinian/israeli conflict in western media: A sentiment analysis experimental study. In *2018 International Conference on Promising Electronic Technologies (ICPET)*, pages 98–103, 2018. doi:10.1109/ICPET.2018.00024.
- [7] M. Alizadeh, M. Kubli, Z. Samei, S. Dehghani, M. Zahedivafa, J. D. Berneo, M. Korobeynikova, and F. Gilardi. Open-source llms for text annotation: a practi-

- cal guide for model setting and fine-tuning. *Journal of Computational Social Science*, 8(1):17, 2025. Epub 2024 Dec 18. doi:10.1007/s42001-024-00345-9.
- [8] Ramy Baly, Giovanni Da San Martino, James Glass, and Preslav Nakov. We can detect your bias: Predicting the political ideology of news articles. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4982–4991, 2020.
- [9] Ramy Baly, Giovanni Da San Martino, James Glass, and Preslav Nakov. We can detect your bias: Predicting the political ideology of news articles. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4982–4991, 2020.
- [10] BBC News. Israel-gaza war: Israeli forces launch assault on rafah, May 2024. Accessed: 2025-08-21. URL: <https://www.bbc.co.uk/news/articles/cjkkj5jejleo>.
- [11] Federico Bianchi, Silvia Terragni, and Dirk Hovy. Bertopic: Neural topic modeling with a class-based tf-idf procedure. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 287–296. Association for Computational Linguistics, 2021. doi:10.18653/v1/2021.emnlp-demo.34.
- [12] Ali Furkan Biten, Lluís Gómez, Marçal Rusiñol, and Dimosthenis Karatzas. Good news, everyone! context driven entity-aware captioning for news images, 2019. URL: <https://arxiv.org/abs/1904.01475>, arXiv:1904.01475.
- [13] David M. Blei, Andrew Y. Ng, and Michael I. Jordan. Latent dirichlet allocation. *Journal of Machine Learning Research*, 3(Jan):993–1022, 2003.
- [14] David M. Blei, Andrew Y. Ng, and Michael I. Jordan. Latent dirichlet allocation. *Journal of Machine Learning Research*, 3(Jan):993–1022, 2003.
- [15] Nitay Calderon, Naveh Porat, Eyal Ben-David, Alexander Chapanin, Zorik Gekhman, Nadav Oved, Vitaly Shalumov, and Roi Reichart. Measuring the robustness of nlp models to domain shifts, 2024. URL: <https://arxiv.org/abs/2306.00168>, arXiv:2306.00168.
- [16] M. Chen, T. Zhang, and S. Wang. Prompting large language models to extract chemical–disease relation precisely and comprehensively at the document level: an

- evaluation study. *PLoS One*, 20(4):e0320123, Apr 2025. doi:10.1371/journal.pone.0320123.
- [17] Francesco Chiossi, Luke Haliburton, Changkun Ou, Andreas Martin Butz, and Albrecht Schmidt. Short-form videos degrade our capacity to retain intentions: Effect of context switching on prospective memory. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI '23, page 1–15. ACM, April 2023. URL: <http://dx.doi.org/10.1145/3544548.3580778>, doi:10.1145/3544548.3580778.
- [18] Kareem Darwish. News consumption in time of conflict: 2021 palestinian-israel war as an example. In *Proceedings of the 14th ACM Web Science Conference 2022*, pages 160–166. ACM, 2022.
- [19] Kareem Darwish et al. Fignews: A shared task on fine-grained news bias and propaganda detection. In *Proceedings of the International Workshop on Fact Extraction and Verification (FEVER)*, 2022. Available at <https://sites.google.com/view/fignews/home>.
- [20] Annelore Deprez and Karin Raeymaeckers. Bias in the news? the representation of palestinians and israelis in the coverage of the first and second intifada. *International Communication Gazette*, 72(1):91–109, 2010. doi:10.1177/1748048509350341.
- [21] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186. Association for Computational Linguistics, 2019.
- [22] Robert M. Entman. Framing bias: Media in the distribution of power. *Journal of Communication*, 57(1):163–173, February 2007. doi:10.1111/j.1460-2466.2006.00336.x.
- [23] Shahira Fahmy. Contrasting visual frames of our times: A framing analysis of english- and arabic-language press coverage of war and terrorism. *International Communication Gazette*, 72(8):695–717, 2010. doi:10.1177/1748048510380801.

- [24] Shahira Fahmy and Daekyung Kim. Picturing the iraq war: Constructing the image of war in the british and us press. *International Communication Gazette*, 70(6):443–462, 2008. doi:10.1177/1748048508096142.
- [25] CFR Education from the Council on Foreign Relations. Timeline for the israel palestine conflict, 2025. URL: <https://education.cfr.org/learn/timeline/israeli-palestinian-conflict-timeline>.
- [26] Johan Galtung and Mari Holmboe Ruge. The structure of foreign news: The presentation of the congo, cuba and cyprus crises in four norwegian newspapers. *Journal of Peace Research*, 2(1):64–91, 1965. doi:10.1177/002234336500200104.
- [27] Matt Gardner, Joel Grus, Mark Neumann, Oyvind Tafjord, Pradeep Dasigi, Nelson Liu, Matthew Peters, Michael Schmitz, and Luke Zettlemoyer. Allennlp: A deep semantic natural language processing platform, 2018. URL: <https://arxiv.org/abs/1803.07640>, arXiv:1803.07640.
- [28] Stephanie Geise and Yi Xu. Effects of visual framing in multimodal media environments: A systematic review of studies between 1979 and 2023. *Journalism & Mass Communication Quarterly*, 0(0):10776990241257586, 0. arXiv:<https://doi.org/10.1177/10776990241257586>, doi:10.1177/10776990241257586.
- [29] Stephanie Geise and Yi Xu. Effects of visual framing in multimodal media environments: A systematic review of studies between 1979 and 2023. *Journalism & Mass Communication Quarterly*, 2024. Advance online publication. doi:10.1177/10776990241257586.
- [30] Ross Girshick. Fast r-cnn, 2015. URL: <https://arxiv.org/abs/1504.08083>, arXiv:1504.08083.
- [31] Motti Golani. 1948: A history of the first arab-israeli war. *Journal of Israeli History*, 29(1):125–129, 2010. doi:10.1080/13531041003595191.
- [32] Tim Groeling. Media bias by the numbers: Challenges and opportunities in the empirical study of partisan news, 2013. doi:10.1146/annurev-polisci-040811-115123.
- [33] Felix Hamborg, Anastasia Zhukova, and Bela Gipp. Automated identification of media bias by word choice and labeling in news articles. In *2019 ACM/IEEE Joint*

- Conference on Digital Libraries (JCDL)*, page 196–205, Champaign, IL, USA, 2019. IEEE. URL: <https://ieeexplore.ieee.org/document/8791197/>, doi:10.1109/JCDL.2019.00036.
- [34] Tony Harcup and Deirdre O’Neill. What is news? news values revisited (again). *Journalism Studies*, 18(12):1470–1488, 2017. doi:10.1080/1461670X.2016.1150193.
- [35] Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. spacy: Industrial-strength natural language processing in python. <https://spacy.io>, 2020. Software package, Version 3.x.
- [36] Clayton J. Hutto and Eric Gilbert. Vader: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media*, 2014. URL: <https://api.semanticscholar.org/CorpusID:12233345>.
- [37] Zahid Ibrar and Musa Khan. Framing and media bias in conflict reporting: A comparative analysis of al-jazeera, cnn, and global times during the 2023 palestine-israel conflict. 2025.
- [38] Avneet Kaur and Arnav Arora. Beyond the battlefield: Framing analysis of media coverage in conflict reporting, 2025. URL: <https://arxiv.org/abs/2506.10421>, arXiv:2506.10421.
- [39] Sungyeon Kim, Xinliang Zhu, Xiaofan Lin, Muhammet Bastan, Douglas Gray, and Suha Kwak. Genius: A generative framework for universal multimodal search, 2025. URL: <https://arxiv.org/abs/2503.19868>, arXiv:2503.19868.
- [40] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation, 2022. URL: <https://arxiv.org/abs/2201.12086>, arXiv:2201.12086.
- [41] Qian Li, Hao Peng, Jianxin Li, Congying Xia, Renyu Yang, Lichao Sun, Philip S. Yu, and Lifang He. A survey on text classification: From traditional to deep learning. *ACM Trans. Intell. Syst. Technol.*, 13(2), April 2022. doi:10.1145/3495162.
- [42] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft coco: Common

- objects in context. In *Computer Vision ECCV-2014*, 2014. URL: https://link.springer.com/chapter/10.1007/978-3-319-10602-1_48, doi:10.1007/978-3-319-10602-1_48.
- [43] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.
- [44] Lefteris Loukas, Ilias Stogiannidis, Prodromos Malakasiotis, and Stavros Vassos. Breaking the bank with chatgpt: Few-shot text classification for finance, 2023. URL: <https://arxiv.org/abs/2308.14634>, arXiv:2308.14634.
- [45] Walid Magdy and Others. Who should set the standards? analysing censored arabic content on facebook during the palestine-israel conflict. *arXiv preprint arXiv:2504.02175*, 2025.
- [46] Omar Shahabudin McDoom. Expert commentary, the israeli-palestinian conflict, and the question of genocide: Prosemitic bias within a scholarly community? *Journal of Genocide Research*, 0(0):1–9, 2024. arXiv:<https://doi.org/10.1080/14623528.2024.2346403>, doi:10.1080/14623528.2024.2346403.
- [47] United Nations. History background. URL: <https://www.un.org/en/situation-in-occupied-palestine-and-israel/history>.
- [48] Douglas L. Nelson, Valerie S. Reed, and John R. Walling. Pictorial superiority effect. *Journal of Experimental Psychology: Human Learning and Memory*, 2(5):523–528, 1976. doi:10.1037/0278-7393.2.5.523.
- [49] Markus Ojala, Mervi Pantti, and Jarkko Kangas. Whose war, whose fault? visual framing of the ukraine conflict in western european newspapers. *International Journal of Communication*, 11(0):25, January 2017.
- [50] Mervi Pantti. Seeing and not seeing the syrian crisis: New visibility and the visual framing of the syrian conflict in seven newspapers and their online editions. *JOMEC Journal*, (3), 2013. doi:10.18573/J.2013.10259.
- [51] Karl Pearson. On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably

- supposed to have arisen from random sampling. *Philosophical Magazine Series 5*, 50(302):157–175, 1900. doi:10.1080/14786440009463897.
- [52] Susan Pedersen. Writing the balfour declaration into the mandate for palestine. *The International History Review*, 45(2):279–291, 2023. arXiv:<https://doi.org/10.1080/07075332.2022.2123377>, doi:10.1080/07075332.2022.2123377.
- [53] Michael Pfau, Michel M. Haigh, Lindsay Logsdon, Christopher Perrine, James P. Baldwin, Rick E. Breitenfeldt, Joel Cesar, Dawn Dearden, Greg Kuntz, Edgar Montalvo, Dwaine Roberts, and Richard Romero. Embedded reporting during the invasion and occupation of iraq: How the embedding of journalists affects television news reports. *Journal of Broadcasting Electronic Media*, 49(4):468–487, 2005. doi:10.1207/s15506878jobem4904_7.
- [54] Bryan A. Plummer, Liwei Wang, Chris M. Cervantes, Juan C. Caicedo, Julia Hockenmaier, and Svetlana Lazebnik. Flickr30k entities: Collecting region-to-phrase correspondences for richer image-to-sentence models, 2016. URL: <https://arxiv.org/abs/1505.04870>, arXiv:1505.04870.
- [55] Thomas E. Powell, Hajo G. Boomgaarden, Knut De Swert, and Claes H. de Vreese. A clearer picture: The contribution of visuals and text to framing effects. *Journal of Communication*, 65(6):997–1017, November 2015. doi:10.1111/jcom.12184.
- [56] Press Gazette. Top world news websites, 2024. Accessed: 2025-08-18.
- [57] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision, 2021. URL: <https://arxiv.org/abs/2103.00020>, arXiv:2103.00020.
- [58] Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. Squad: 100,000+ questions for machine comprehension of text, 2016. URL: <https://arxiv.org/abs/1606.05250>, arXiv:1606.05250.
- [59] S. Raza, D. J. Reji, and C. Ding. Dbias: detecting biases and ensuring fairness in news articles. *International Journal of Data Science and Analytics*, pages 1–21, September 2022. Epub ahead of print. doi:10.1007/s41060-022-00359-4.

- [60] Marta Recasens, Cristian Danescu-Niculescu-Mizil, and Dan Jurafsky. Linguistic models for analyzing and detecting biased language. In Hinrich Schuetze, Pascale Fung, and Massimo Poesio, editors, *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1650–1659, Sofia, Bulgaria, August 2013. Association for Computational Linguistics. URL: <https://aclanthology.org/P13-1162/>.
- [61] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection, 2016. URL: <https://arxiv.org/abs/1506.02640>, arXiv:1506.02640.
- [62] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter, 2020. URL: <https://arxiv.org/abs/1910.01108>, arXiv:1910.01108.
- [63] Carol B. Schwalbe. Remembering our shared past: Visually framing the iraq war on u.s. news websites. *Journal of Computer-Mediated Communication*, 12(1):264–289, oct 2006. doi:10.1111/j.1083-6101.2006.00325.x.
- [64] Carol B. Schwalbe. Visually framing the invasion and occupation of iraq in *TIME*, *Newsweek*, and *U.S. News & World Report*. *International Journal of Communication*, 7:1176–1201, 2013. URL: <https://ijoc.org/index.php/ijoc/article/view/924>.
- [65] Scrapy developers. Scrapy: An open source web crawling framework for python. <https://scrapy.org/>, 2025. Accessed: 19 Aug 2025. URL: <https://scrapy.org/>.
- [66] Krishna Kumar Singh, Dhruv Mahajan, Kristen Grauman, Yong Jae Lee, Matt Feiszli, and Deepti Ghadiyaram. Don’t judge an object by its context: Learning to overcome contextual bias, 2020. URL: <https://arxiv.org/abs/2001.03152>, arXiv:2001.03152.
- [67] SK. The balfour declaration from the perspective of the palestinian people, lecture by prof. rashid khalidi, columbia university, 2017. URL: <https://www.un.org/unispal/document/lecture-rashid-khalidi-balfour-declaration-2nov2017/>.

- [68] Sophie. Why is the media so biased against the palestinians?, March 2024. URL: <https://socialistworker.co.uk/in-depth/why-is-the-media-so-biased-against-the-palestinians/>.
- [69] Tanja Spinde, Thomas Pilz, Gabriella Lapesa, Peter Fankhauser, and Steffen Eger. Detecting lexical semantic change in language models with contextualized embeddings. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL 2021)*, pages 7041–7053, 2021. URL: <https://aclanthology.org/2021.acl-long.546>.
- [70] Timo Spinde, Smilla Hinterreiter, Fabian Haak, Terry Ruas, Helge Giese, Norman Meuschke, and Bela Gipp. The media bias taxonomy: A systematic literature review on the forms and automated detection of media bias. *ACM Computing Surveys [in Review]*, 2023.
- [71] Timo Spinde, Lada Rudnitskaia, Felix Hamborg, and Bela Gipp. *Identification of Biased Terms in News Articles by Comparison of Outlet-Specific Word Embeddings*, page 215–224. Springer International Publishing, 2021. URL: http://dx.doi.org/10.1007/978-3-030-71305-8_17, doi: 10.1007/978-3-030-71305-8_17.
- [72] Kate Starbird, Ahmer Arif, and Tom Wilson. Disinformation as collaborative work: Surfacing the participatory nature of strategic information operations. *Proc. ACM Hum.-Comput. Interact.*, 3(CSCW), November 2019. doi:10.1145/3359229.
- [73] Mark Tessler. *A history of the Israeli-Palestinian conflict*. Indiana University Press, 2009.
- [74] The Guardian. Hamas accepts us ceasefire plan but says it does not meet demands to end war in gaza, May 2025. Accessed: 2025-08-21. URL: <https://www.theguardian.com/world/2025/may/30/hamas-us-ceasefire-plan-accepted-israel-does-not-meet-demands-end-war-gaza>
- [75] The Washington Institute for Near East Policy. Gaza war media coverage dataset, 2024. Database of 436 news stories from seven major U.S. outlets, October 2023–April 2024. URL: <https://www.washingtoninstitute.org/gaza-war-media-database>.

- [76] Alasdair Tran, Alexander Mathews, and Lexing Xie. Transform and tell: Entity-aware news image captioning, 2020. URL: <https://arxiv.org/abs/2004.08070>, arXiv:2004.08070.
- [77] United Nations Office for the Coordination of Humanitarian Affairs (OCHA) - occupied Palestinian territory. Data on casualties, 2025. URL: <https://www.ochaopt.org/data/casualties>.
- [78] Uppsala Conflict Data Program (UCDP). Ucdp georeferenced event dataset (ged), global, 2025. Version: [25.1]; Accessed: 2025-08-16. URL: <https://ucdp.uu.se/downloads/>.
- [79] Sebastian Usher. Gaza ceasefire once again in doubt as first phase nears end. *BBC News*, February 2025. Reporting from Jerusalem. URL: <https://www.bbc.co.uk/news/articles/c8x4pxwnelyo>.
- [80] Ramakrishna Vedantam, C. Lawrence Zitnick, and Devi Parikh. Cider: Consensus-based image description evaluation. In *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, page 4566–4575, Boston, MA, USA, 2015. IEEE. URL: <http://ieeexplore.ieee.org/document/7299087/>, doi: 10.1109/CVPR.2015.7299087.
- [81] Anna Verbytska. Topic modelling as a method for framing analysis of news coverage of the russia-ukraine war in 2022–2023. *Language Communication*, 99:174–193, 2024. doi:10.1016/j.langcom.2024.10.004.
- [82] Madhukar Reddy Vongala, Saurabh Srivastava, and Jana Košecká. Compositional image-text matching and retrieval by grounding entities, 2025. URL: <https://arxiv.org/abs/2505.02278>, arXiv:2505.02278.
- [83] Vikas Yadav and Steven Bethard. A survey on recent advances in named entity recognition from deep learning models. In Emily M. Bender, Leon Derczynski, and Pierre Isabelle, editors, *Proceedings of the 27th International Conference on Computational Linguistics*, page 2145–2158, Santa Fe, New Mexico, USA, August 2018. Association for Computational Linguistics. URL: <https://aclanthology.org/C18-1182/>.
- [84] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng

Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025. URL: <https://arxiv.org/abs/2505.09388>, arXiv:2505.09388.

Appendix A

First appendix

A.1 Code Repository

A cleaned code repository archive was uploaded with this report. Since the image and article dataset was huge, it could not be uploaded as a zip file. The source file for the LaTeX report (zip) is added to the same project archive file.

The complete code repo and the datasets are present on the University of Edinburgh SMASH server at the location (documented in the archive Readme.md as well):

```
SMASH1 server (smash1.inf.ed.ac.uk) in the location:  
.  
mnt/raid1/MSCs/AY2425/skaul/diss_crawler
```

All final data_assets created are present at:

```
/mnt/raid1/MSCs/AY2425/skaul/diss_crawler/notebooks/data_assets/
```

Raw data files stored in the location:

```
/mnt/raid1/MSCs/AY2425/skaul/diss_crawler/notebooks/data/
```

Final scraped and downloaded images stored at the location:

```
/mnt/raid1/MSCs/AY2425/skaul/diss_crawler/notebooks/img/
```

A.2 Data Collection JSON Format

```
{
```

```

"title": "Airstrikes reported in northern Gaza",
"url": "https://www.example.com/world/2024/05/13/...",
"source_domain": "cnn.com",
"date_published": "2024-05-13T18:42:00Z",
"authors": ["Reporter Name"],
"text": "....",
"matched_keywords": ["Gaza", "Israeli"],
"images": [
  {
    "src": "https://media.example.com/hero.jpg",
    "role": "hero",
    "caption": "Smoke rises over northern Gaza after strikes.",
    "credit": "Getty Images"
  },
  {
    "src": "https://media.example.com/inline1.jpg",
    "role": "inline",
    "caption": null,
    "credit": null
  }
]
}

```

A.3 Classification Prompt Template

```

"You are a precise news classification assistant. "
"Classify each article into exactly ONE of the allowed labels and "
"return a JSON object that matches the schema. No extra fields."
SCHEMA = {
  "type": "json_schema",
  "json_schema": {
    "name": "news_classification_batch",
    "strict": True,
    "schema": {
      "type": "object",

```

```

    "properties": {
      "results": {
        "type": "array",
        "items": {
          "type": "object",
          "properties": {
            "label": {"type": "string", "enum": LABELS}
          },
          "required": ["label"],
          "additionalProperties": False
        }
      }
    },
    "required": ["results"],
    "additionalProperties": False
  }
}

"Labels:",
  "1) Conflict-related { Direct news on Israel{Palestine conflict",

  "2) Political Statements { Statements by
  leaders/officials on the conflict",
  "3) Protests and Public Reactions { Citizen protests
  or opinion pieces",
  "4) Irrelevant { Not related to the conflict",
  "",
  "Return JSON ONLY with this shape:
  {\"results\": [{\"label\": \"<one of allowed>\"}, ...]}\",

```


A.4 NER Entity List

Table A.1: Comprehensive lexicons of entities associated with Israel and Palestine, extracted via spaCy NER and curated manually.

Side	Category	Entities
Israel	Core Terms	israel, israeli, israelis, zionist, zionists
	Military & Security	idf, israel defense forces, israeli defense forces, tzahal, shin bet, mossad, aman, israeli military, israeli army, israeli forces, israeli soldiers, israeli troops
	Political Leaders & Parties	netanyahu, benjamin netanyahu, binyamin netanyahu, bibi, gantz, benny gantz, lapid, yair lapid, bennett, naftali bennett, gallant, yoav gallant, smotrich, ben gvir, itamar ben gvir, likud, blue and white, kahol lavan, yesh atid, yamina, knesset, israeli parliament, israeli government
	Cities & Regions	jerusalem, west jerusalem, east jerusalem, jewish quarter, tel aviv, haifa, ashdod, beersheba, eilat, nazareth, petah tikva, rishon lezion, rehovot, bat yam, holon, bnei brak, ramat gan, ashkelon, netanya, kiryat gat, sderot, ofakim, dimona, galilee, negev, golan heights, judea, samaria
	Communities & Settlements	kibbutz, kibbutzim, moshav, settler, settlers, israeli settlement, israeli settlements

Table A.2: Comprehensive lexicons of entities associated with Israel and Palestine, extracted via spaCy NER and curated manually.

Side	Category	Entities
Palestine	Core Terms	palestine, palestinian, palestinians, state of palestine, gaza, gaza strip, gazans, gazan, gaza city, west bank, westbank, occupied west bank
	Cities & Regions	rafah, jabalia, jabalya, nuseirat, deir al balah, deir al-balah, khan younis, khan yunis, beit hanoun, beit lahia, northern gaza, southern gaza, central gaza, ramallah, hebron, al khalil, bethlehem, beit lahm, nablus, jenin, tulkarm, qalqilya, salfit, jericho, ariha, east jerusalem, arab jerusalem
	Political Orgs	hamas, harakat al-muqawama al-islamiyya, islamic resistance movement, fatah, harakat al-tahrir al-watani al-filastini, plo, palestine liberation organization, pa, palestinian authority, palestinian national authority, pflp, popular front, dflp, democratic front
	Military Wings & Factions	qassam, al-qassam, izz ad-din al-qassam, qassam brigades, aqsa martyrs, al-aqsa martyrs brigades, al aqsa martyrs, saraya al-quds, quds brigades
	Leaders	abbas, mahmoud abbas, abu mazen, haniyeh, ismail haniyeh, sinwar, yahya sinwar, meshaal, khaled meshaal, barghouti, marwan barghouti, erekat, saeb erekat
	Hospitals & Institutions	shifa, al-shifa, al shifa hospital, shifa hospital, al-quds hospital, indonesian hospital, european hospital

A.5 BERT and LDA Results

Table A.3: Illustrative topics extracted using Latent Dirichlet Allocation (LDA) on conflict-related text. Topics reflect broader thematic clusters rather than fine-grained actor mentions.

Topic	Top Terms (examples)
Conflict	war, attack, missile, strike, ceasefire, fighting, rocket, killed, forces
Politics	government, minister, negotiations, agreement, peace, authority, parliament, law
International Response	united nations, us, eu, sanctions, diplomacy, security council, aid
Humanitarian	refugees, food, hospital, humanitarian, displaced, water, crisis, shelter

Table A.4: Illustrative entities extracted using transformer-based NER and BERT-guided classification. Entities are grouped into curated lexicons for Israel and Palestine.

Side	Category	Entities (examples)
Israel	Military	idf, israel defense forces, israeli army, israeli soldiers, mossad, shin bet
	Political Leaders	netanyahu, gantz, lapid, bennett, gallant, smotrich, ben gvir, likud, knesset
	Cities & Regions	jerusalem, tel aviv, haifa, sderot, beersheba, golan heights, galilee, negev
Palestine	Political Orgs	hamas, islamic jihad, fatah, plo, pa, pflp, dflp, quds brigades
	Leaders	abbas, haniyeh, sinwar, meshaal, barghouti, erekat
	Cities & Regions	gaza, rafah, khan younis, ramallah, hebron, nablus, jenin, bethlehem

A.6 Scene Detection Classification Evaluation

Table A.5: Confusion matrix on the test set (binary classifier).

	Actual Positive	Actual Negative
Predicted Positive	65 (TP)	9 (FP)
Predicted Negative	8 (FN)	18 (TN)
<i>Precision = 0.878, Recall = 0.890, F1 = 0.884.</i>		

Table A.6: Classification report on the test set (Military vs Non-Military).

Class	Precision	Recall	F1-score	Support
Non-Military	0.88	0.92	0.90	48
Military	0.92	0.88	0.90	52
Accuracy		0.90		100
Macro avg	0.90	0.90	0.90	100
Weighted avg	0.90	0.90	0.90	100

Table A.7: Classification report on the test set (Casualty vs Non-Casualty).

Class	Precision	Recall	F1-score	Support
Non-Casualty	0.94	0.96	0.95	53
Casualty	0.96	0.94	0.95	47
Accuracy		0.95		100
Macro avg	0.95	0.95	0.95	100
Weighted avg	0.95	0.95	0.95	100

A.7 CLIP Prompt Templates

Table A.8: Exact CLIP prompt templates used (checkpoint: openai/clip-vit-base-patch32). Positive prompts are used for scoring; negatives are logged as sanity checks.

Class	Type	Prompt text
Victim/Casualty	Positive	a photo showing injured civilians
Victim/Casualty	Positive	a photo showing wounded people
Victim/Casualty	Positive	a photo showing dead bodies covered with sheets
Victim/Casualty	Positive	a photo of a funeral with grieving families
Victim/Casualty	Positive	a photo of an ambulance with injured people
Victim/Casualty	Positive	a photo inside a hospital treating the wounded
Victim/Casualty	Positive	a photo of bloodstains and first aid being given
Victim/Casualty	Positive	a photo of people mourning casualties
Victim/Casualty	Negative	a normal scene without any casualties or injured people
Military	Positive	a photo of armed personnel military or militants or military vehicles
Military	Negative	a photo without any military content

A.8 Side Attribution Prompt Templates

Task	Prompt (verbatim)
Destruction → Side (receiver)	You are a classification system. Task: For each caption below, determine who was the receiver of the destruction. Rules: - Only output "Israel" or "Palestine" as the receiver. - If unclear, output null. - Do NOT output explanations or extra text. Return: A JSON array where each element matches this structure: "receiver": "Israel" or "receiver": "Palestine" or "receiver": null
Victim → Side (victim nationality)	Captions: 1. ... (batched list, one per line) You are labelling news image captions from the Israel-Palestine conflict. Your task is to identify the nationality of the VICTIM(S) shown in the image described by the caption. Do NOT label based on who is present unless they are the victims. Ignore the nationality of the perpetrator or attacker. If no victim is shown, output "None". Instructions: - Output only one word: "Palestine", "Israel", or "None". - Read the entire caption before deciding. - If the victim is an Israeli hostage, funeral attendee for an Israeli, or Israeli killed/injured person -> output "Israel". - If the victim is a Palestinian civilian, detainee, injured, dead, or displaced person -> output "Palestine". - If no clear victim is mentioned -> output "None". You will receive MULTIPLE captions. For each caption, produce exactly one classification. Return a JSON array of objects, one per caption, in the SAME order, each exactly like: "receiver": "Palestine" OR "receiver": "Israel" OR "receiver": "None" Captions: 1. ... (batched list, one per line)

Table A.9: Prompts used for GPT-3.5 Turbo: Destruction and Victim attribution.

Task	Prompt (verbatim)
Military → Affiliation (system + user)	System prompt: You label news image captions by MILITARY AFFILIATION mentioned/depicted. Output one of exactly: israeli, palestinian, mixed, unclear. Definitions: - israeli: caption clearly refers to Israeli military/security/forces (e.g., IDF, Israeli soldiers/police) as the military actor shown or discussed. - palestinian: caption clearly refers to Palestinian armed/security groups (e.g., Hamas, Palestinian forces/security) as the military actor shown or discussed. - mixed: both sides' military/armed/security are clearly present/referred to together as actors. - unclear: no clear military/armed/security actor; civilians, politicians, protests, casualties without explicit military affiliation, or ambiguous. Read the entire caption; decide based on who the MILITARY or SECURITY actor(s) are, not victims. Return JSON only---no extra text. User prompt: You will receive MULTIPLE captions. For each caption, output exactly one object with this schema: "military_affiliation":"israeli palestinian mixed unclear" Return a JSON array of the same length, in order. Captions: 1. ... (batched list, one per line)

Table A.10: Prompts used for GPT-3.5 Turbo: Military affiliation classification (system and user prompts).

A.9 Significance Tests Extended Results

Source	Palestine %	N	χ^2	p-value	Significant
dailymail	25.0	2566	1078.15	$1.856860 \times 10^{-236}$	Yes
foxnews	30.2	937	667.60	$3.315414 \times 10^{-147}$	Yes
apnews	31.2	634	493.91	$2.005484 \times 10^{-109}$	Yes
nypost	30.0	620	433.82	2.396625×10^{-96}	Yes
express	31.3	355	277.44	2.715083×10^{-62}	Yes
theguardian	26.4	535	261.97	6.371054×10^{-59}	Yes
news18	32.1	168	141.23	1.434457×10^{-32}	Yes
newsweek	26.9	253	130.92	2.576034×10^{-30}	Yes
cnn	26.7	232	118.14	1.613820×10^{-27}	Yes
independent	20.6	321	74.70	5.467957×10^{-18}	Yes
bbc	21.4	238	63.12	1.949456×10^{-15}	Yes
usatoday	31.8	44	36.01	1.960882×10^{-9}	Yes
india	40.0	20	29.36	6.013373×10^{-8}	Yes
abcnews.go	20.2	124	27.09	1.939101×10^{-7}	Yes
indianexpress	17.5	137	18.58	1.627365×10^{-5}	Yes
hindustantimes	15.5	206	17.79	2.468685×10^{-5}	Yes
cnnbc	17.4	46	6.08	1.368007×10^{-2}	Yes

Table A.11: Full significance table for military portrayal (deviation from **GED baseline: Palestine = 92.6%**).

Source	Palestine %	N	χ^2	p-value	Significant
dailymail	73.3	2531	4712.02	0.000000	Yes
nypost	62.7	608	2372.31	0.000000	Yes
foxnews	53.8	208	1283.52	$4.316494 \times 10^{-281}$	Yes
apnews	80.7	1340	1174.81	$1.816894 \times 10^{-257}$	Yes
bbc	79.0	414	442.31	3.406727×10^{-98}	Yes
theguardian	86.7	1032	355.70	2.435969×10^{-79}	Yes
cnn	81.3	315	256.18	1.165077×10^{-57}	Yes
hindustantimes	75.2	153	240.65	2.835109×10^{-54}	Yes
newsweek	75.7	140	209.18	2.069481×10^{-47}	Yes
express	81.6	207	160.42	9.172651×10^{-37}	Yes
independent	86.4	361	131.82	1.634092×10^{-30}	Yes
cnbc	52.6	19	123.95	8.627339×10^{-29}	Yes
usatoday	76.5	85	118.10	1.649698×10^{-27}	Yes
news18	82.2	90	64.55	9.424562×10^{-16}	Yes
abcnews.go	87.4	135	40.50	1.969639×10^{-10}	Yes
indianexpress	87.2	109	34.45	4.368151×10^{-9}	Yes
india	80.0	10	9.51	2.045810×10^{-3}	Yes

Table A.12: Full significance table for victim portrayal (deviation from **OCHA Baseline**), sorted by ascending p-value.

Source	Gaza %	N	χ^2	p-value	Significant
dailymail	71.9	2680	1574.57	0.000000	Yes
foxnews	55.8	312	585.84	$2.011609 \times 10^{-129}$	Yes
nypost	58.8	369	582.41	$1.123187 \times 10^{-128}$	Yes
apnews	81.5	1055	172.61	1.988734×10^{-39}	Yes
newsweek	74.8	266	114.46	1.031340×10^{-26}	Yes
express	72.4	203	112.96	2.206684×10^{-26}	Yes
independent	80.5	431	84.29	4.271064×10^{-20}	Yes
news18	74.6	134	58.89	1.667607×10^{-14}	Yes
hindustantimes	81.0	284	51.15	8.547261×10^{-13}	Yes
theguardian	86.1	934	50.82	1.011816×10^{-12}	Yes
bbc	80.7	254	48.02	4.222193×10^{-12}	Yes
cnn	82.5	292	39.18	3.858098×10^{-10}	Yes
india	68.0	25	20.77	5.175733×10^{-6}	Yes
indianexpress	85.8	155	9.20	2.425320×10^{-3}	Yes
usatoday	84.3	70	6.33	1.189744×10^{-2}	Yes
abcnews.go	87.2	172	6.27	1.226760×10^{-2}	Yes
cnbc	84.6	39	3.24	7.183726×10^{-2}	No

Table A.13: Full significance table for destruction portrayal (deviation from **GED baseline: Gaza = 92.3%**), sorted by ascending p-value.

A.10 Real-world Baselines

A.10.1 GED - Geo Referenced event Dataset

Disaggregated Datasets

UCDP Georeferenced Event Dataset (GED) Global version 25.1

This dataset is UCDP's most disaggregated dataset, covering individual events of organized violence (phenomena of lethal violence occurring at a given time and place). These events are sufficiently fine-grained to be geo-coded down to the level of individual villages, with temporal durations disaggregated to single, individual days.

Available as:



Please cite:

- Davies, S., Pettersson, T., Sollenberg, M., & Öberg, M. (2025). Organized violence 1989–2024, and the challenges of identifying civilian victims. *Journal of Peace Research*, 62(4).
- Sundberg, Ralph and Erik Melander (2013) Introducing the UCDP Georeferenced Event Dataset. *Journal of Peace Research* 50(4).

Figure A.1: UCDP Geo-reference dataset used

A.10.2 UN OCHA casualty dashboard

time, area, context and affiliation of the casualties.

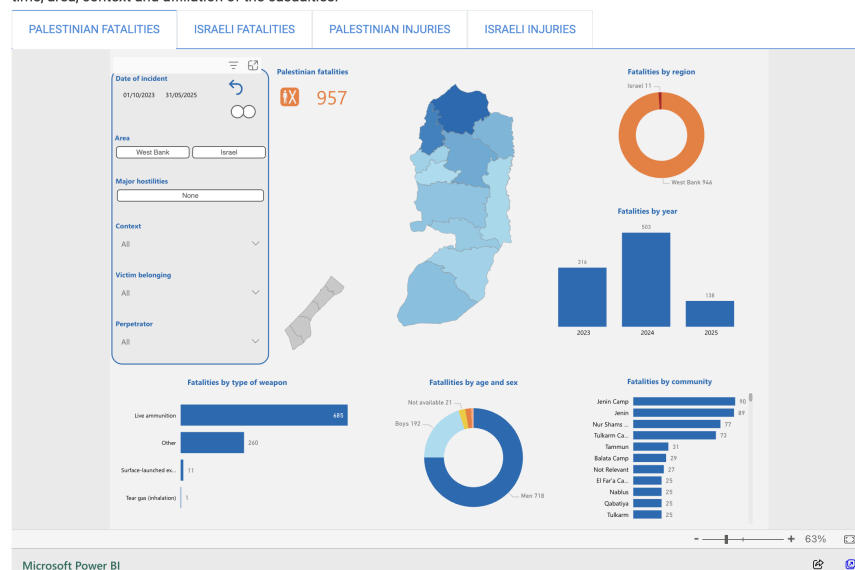


Figure A.2: UN OCHA Dashboard which was used to extract metrics of Israeli vs Palestinian Casualties (injuries+fatalities)